```
// 文件: Assets/Scripts/Game/LoadingScreen.cs
using System.Collections;
using UnityEngine;
using UnityEngine.UI;
namespace Rescue
{
    /// <summary>
    /// 启动页：和微信封面同一张图（tools/build_loading_art.py 合成，标题、进度槽、健康忠告、适龄标都在图上），
    /// 这里只往图上的槽里画金色进度和一行状态字。云存档回来并且最短展示时间到了才进主界面。
    /// </summary>
    public sealed class LoadingScreen : MonoBehaviour
    {
        public const float MinSeconds = 2.6f;
        /// <summary>图是 1080x1920，槽的位置和 build_loading_art.py 一致（720 设计尺寸 ×1.5）。</summary>
        const float ArtW = 1080f, ArtH = 1920f;
        const float BarW = 600f, BarH = 15f, BarBottom = 321f;
        const float TextBottom = 348f;
        RectTransform _root, _frame, _stage;
        CanvasGroup _group;
        RectTransform _fill, _head;
        Text _status;
        float _born;
        bool _hold;
        public static LoadingScreen Open(Transform host)
        {
            var go = new GameObject("Loading");
            go.transform.SetParent(host, false);
            var s = go.AddComponent<LoadingScreen>();
            s.Build();
            return s;
        }
        void Build()
        {
            _born = Time.realtimeSinceStartup;
            _root = Ui.Rect(Ui.Root, "Loading");
            _root.anchorMin = Vector2.zero;
            _root.anchorMax = Vector2.one;
            _root.offsetMin = _root.offsetMax = Vector2.zero;
            _group = _root.gameObject.AddComponent<CanvasGroup>();
            _group.blocksRaycasts = true;
            var night = Ui.Rect(_root, "night");
            Ui.PinBleed(night);
            var nightImg = night.gameObject.AddComponent<Image>();
            nightImg.color = Pal.Hex("140D09");
            nightImg.raycastTarget = true;
            // 整张图等比铺满（和微信 noBorder 一样），不拉伸；长屏裁左右，要紧的东西都在中间。
            _frame = Ui.Rect(_root, "frame");
            Ui.PinBleed(_frame);
            _stage = Ui.Rect(_frame, "art");
            _stage.anchorMin = _stage.anchorMax = new Vector2(0.5f, 0.5f);
            _stage.sizeDelta = new Vector2(ArtW, ArtH);
            var img = _stage.gameObject.AddComponent<Image>();
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        img.sprite = Splash();
        img.raycastTarget = false;
        if (img.sprite == null) img.color = Pal.Hex("140D09");
        BuildBar();
        Fit();
        _root.SetAsLastSibling();
    }
    /// <summary>金色进度落在图上画好的黑漆槽里，前端一颗亮点；上面一行状态字。</summary>
    void BuildBar()
    {
        var track = Ui.Box(_stage, new Vector2(BarW, BarH), Vector2.zero, Pal.Hex("000000").A(0.45f), Art.Rounded, "barTrack");
        DockBottom(track.rectTransform, BarBottom, BarH);
        var fill = Ui.Box(track.transform, new Vector2(BarH, BarH), Vector2.zero, Pal.Hex("E6B54A"), Art.Rounded, "barFill");
        _fill = fill.rectTransform;
        _fill.anchorMin = _fill.anchorMax = new Vector2(0f, 0.5f);
        _fill.pivot = new Vector2(0f, 0.5f);
        _fill.anchoredPosition = Vector2.zero;
        var shine = Ui.Box(_fill, new Vector2(10, BarH * 0.32f), Vector2.zero, Color.white.A(0.35f), Art.Rounded, "shine");
        shine.rectTransform.anchorMin = new Vector2(0f, 0.5f);
        shine.rectTransform.anchorMax = new Vector2(1f, 0.5f);
        shine.rectTransform.sizeDelta = new Vector2(-BarH * 0.6f, BarH * 0.32f);
        shine.rectTransform.anchoredPosition = new Vector2(0f, BarH * 0.16f);
        var head = Ui.Box(_fill, new Vector2(BarH * 3.2f, BarH * 3.2f), Vector2.zero, Pal.Hex("FFE7A8").A(0.8f), Art.SoftDot, "head"
            );
        _head = head.rectTransform;
        _head.anchorMin = _head.anchorMax = new Vector2(1f, 0.5f);
        _head.anchoredPosition = new Vector2(-BarH * 0.4f, 0f);
        _status = Ui.Label(_stage, "正在加载资源", 26, Pal.Hex("EBDDBE"), Vector2.zero, new Vector2(BarW, 34), TextAnchor.Mid
            dleCenter, null, "status");
        DockBottom(_status.rectTransform, TextBottom, 34);
        foreach (var bar in new[] { track, fill, shine }) bar.pixelsPerUnitMultiplier = 18f / (BarH * 0.5f);
        shine.pixelsPerUnitMultiplier *= 2f;
        ApplyFill(0f);
    }
    void Fit()
    {
        if (_frame == null || _stage == null) return;
        var size = _frame.rect.size;
        if (size.x < 1f || size.y < 1f) return;
        float k = Mathf.Max(size.x / ArtW, size.y / ArtH);
        _stage.localScale = new Vector3(k, k, 1f);
    }
    public void SetFill(float p)
    {
        _hold = true;
        ApplyFill(p);
    }
    void ApplyFill(float p)
    {
        if (_fill == null) return;
        p = Mathf.Clamp01(p);
        _fill.sizeDelta = new Vector2(Mathf.Lerp(BarH, BarW, p), BarH);
```

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        if (_status != null) _status.text = p >= 0.999f ? "开始游戏" : $"正在加载资源  {Mathf.RoundToInt(p * 100)}%";
    }
    /// <summary>手写的 meta 若没导成 Sprite，退回整张贴图，避免铺成一块白圆角。</summary>
    static Sprite Splash()
    {
        var sprite = Resources.Load<Sprite>("Art/loading");
        if (sprite != null && sprite.texture != null && sprite.texture.width > 128) return sprite;
        var tex = Resources.Load<Texture2D>("Art/loading");
        if (tex == null || tex.width < 128) return null;
        return Sprite.Create(tex, new Rect(0, 0, tex.width, tex.height), new Vector2(0.5f, 0.5f), 100f, 0, SpriteMeshType.FullRect);
    }
    static void DockBottom(RectTransform rt, float y, float h)
    {
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0f);
        rt.pivot = new Vector2(0.5f, 0f);
        rt.sizeDelta = new Vector2(rt.sizeDelta.x, h);
        rt.anchoredPosition = new Vector2(0f, y);
    }
    void Update()
    {
        Fit();
        if (_head != null)
        {
            float pulse = 0.85f + 0.15f * Mathf.Sin(Time.realtimeSinceStartup * 9f);
            _head.localScale = new Vector3(pulse, pulse, 1f);
        }
        if (_hold || _fill == null) return;
        float p = Mathf.Clamp01((Time.realtimeSinceStartup - _born) / MinSeconds);
        ApplyFill(Mathf.Lerp(0.06f, 0.9f, p));
    }
    public void BringToFront()
    {
        if (_root) _root.SetAsLastSibling();
    }
    public IEnumerator WaitMin()
    {
        while (Time.realtimeSinceStartup - _born < MinSeconds) yield return null;
    }
    public IEnumerator FadeOut()
    {
        float t = 0f;
        const float dur = 0.4f;
        while (t < dur)
        {
            t += Time.unscaledDeltaTime;
            if (_group) _group.alpha = 1f - Mathf.Clamp01(t / dur);
            yield return null;
        }
        Destroy(gameObject);
    }
    void OnDestroy()
    {
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            if (_root) Destroy(_root.gameObject);
        }
    }
}
// 文件: Assets/Scripts/Game/HomeScreen.cs
using System;
using System.Collections.Generic;
using Rescue.Sim;
using UnityEngine;
using UnityEngine.UI;
namespace Rescue
{
    /// <summary>神秘商人（孙二娘）的货：主动道具战斗里拖上战场放出，被动道具开局就生效。</summary>
    public sealed class ShopItem
    {
        public int Id;
        public string Name, Desc, Rarity, Icon;
        public bool Active;
        public int Price;
        public Func<int> Count;
        public Action Give, Take;
        public bool Owned() => Count() >= (Id == 8 ? 2 : 1);
        public static readonly string[] Rarities = { "稀有", "卓越", "史诗", "传说" };
        /// <summary>稀有绿、卓越蓝、史诗紫、传说橙，和圆底同色系。</summary>
        public static Color ColorOf(string rarity) => rarity switch
        {
            "传说" => Pal.Hex("F08A24"),
            "史诗" => Pal.Hex("B45CE0"),
            "卓越" => Pal.Hex("3FA9E0"),
            _ => Pal.Hex("6FBF3A"),
        };
        public Color RarityColor => ColorOf(Rarity);
        public Color DiscColor => Rarity switch
        {
            "传说" => Pal.Hex("F6C46B"),
            "史诗" => Pal.Hex("C9A6EE"),
            "卓越" => Pal.Hex("A4D8F2"),
            _ => Pal.Hex("C5E08A"),
        };
        /// <summary>抽奖权重：越稀有越难出。</summary>
        public int Weight => Rarity switch { "传说" => 12, "史诗" => 22, "卓越" => 30, _ => 36 };
        static int B(bool v) => v ? 1 : 0;
        public static readonly ShopItem[] All =
        {
            new ShopItem { Id = 0, Name = "生辰纲", Desc = "开局炊饼 +30，抢先聚义", Rarity = "卓越", Icon = "it_gang", Price =
                Economy.ItemPrices[0], Count = () => B(Save.D.gang), Give = () => { Save.D.gang = true; Save.NotePassive(0); }, Take =
                () => { Save.D.gang = false; Save.NotePassive(0); } },
            new ShopItem { Id = 1, Name = "替天行道旗", Desc = "全军伤害 +22%，全局生效", Rarity = "史诗", Icon = "it_flag", Pri
                ce = Economy.ItemPrices[1], Count = () => B(Save.D.flag), Give = () => { Save.D.flag = true; Save.NotePassive(1); }, Tak
                e = () => { Save.D.flag = false; Save.NotePassive(1); } },
            new ShopItem { Id = 2, Name = "水泊", Desc = "官兵移速 -25%，全局生效", Rarity = "史诗", Icon = "it_marsh", Price =
                Economy.ItemPrices[2], Count = () => B(Save.D.marsh), Give = () => { Save.D.marsh = true; Save.NotePassive(2); }, Take
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            = () => { Save.D.marsh = false; Save.NotePassive(2); } },
        new ShopItem { Id = 3, Name = "石碣", Desc = "官兵靠近宋江时，天降石碣砸下", Rarity = "传说", Icon = "it_stele", Pric
            e = Economy.ItemPrices[3], Count = () => B(Save.D.stele), Give = () => { Save.D.stele = true; Save.NotePassive(3); }, Tak
            e = () => { Save.D.stele = false; Save.NotePassive(3); } },
        new ShopItem { Id = 4, Name = "蒙汗药", Desc = "全场官兵昏睡 3 秒", Rarity = "史诗", Icon = "it_menghan", Active = tr
            ue, Price = Economy.ItemPrices[4], Count = () => Save.D.menghan, Give = () => { Save.D.menghan++; Save.NoteActive(
            4); }, Take = () => { Save.D.menghan = Mathf.Max(0, Save.D.menghan - 1); Save.NoteActive(4); } },
        new ShopItem { Id = 5, Name = "天书", Desc = "拖到哪个兵身上，哪个升一级", Rarity = "传说", Icon = "it_tianshu", Act
            ive = true, Price = Economy.ItemPrices[5], Count = () => Save.D.tianshu, Give = () => { Save.D.tianshu++; Save.NoteAct
            ive(5); }, Take = () => { Save.D.tianshu = Mathf.Max(0, Save.D.tianshu - 1); Save.NoteActive(5); } },
        new ShopItem { Id = 6, Name = "孙二娘包子", Desc = "五成半回一颗心，否则掉一颗", Rarity = "卓越", Icon = "bun", Ac
            tive = true, Price = Economy.ItemPrices[6], Count = () => Save.D.baozi, Give = () => { Save.D.baozi++; Save.NoteActive
            (6); }, Take = () => { Save.D.baozi = Mathf.Max(0, Save.D.baozi - 1); Save.NoteActive(6); } },
        new ShopItem { Id = 7, Name = "续命丹", Desc = "宋江多八颗心，能硬吃第一波", Rarity = "传说", Icon = "it_xuming",
            Price = Economy.ItemPrices[7], Count = () => B(Save.D.xuming), Give = () => { Save.D.xuming = true; Save.NotePassive(
            7); }, Take = () => { Save.D.xuming = false; Save.NotePassive(7); } },
        new ShopItem { Id = 8, Name = "方便铲", Desc = "开局送 2 把铲子，多开两块地", Rarity = "稀有", Icon = "shovel", Price
             = Economy.ItemPrices[8], Count = () => Save.D.shovels, Give = () => { Save.D.shovels = 2; Save.NotePassive(8); }, Take
            = () => { Save.D.shovels = 0; Save.NotePassive(8); } },
        new ShopItem { Id = 9, Name = "攻速符", Desc = "单位攻速 +40%，持续十二秒", Rarity = "史诗", Icon = "it_fu", Active
            = true, Price = Economy.ItemPrices[9], Count = () => Save.D.fu, Give = () => { Save.D.fu++; Save.NoteActive(9); }, Take
            = () => { Save.D.fu = Mathf.Max(0, Save.D.fu - 1); Save.NoteActive(9); } },
        new ShopItem { Id = 10, Name = "御敌千里", Desc = "远程单位攻击范围翻倍，全局生效", Rarity = "稀有", Icon = "it_fa
            r", Price = Economy.ItemPrices[10], Count = () => B(Save.D.far), Give = () => { Save.D.far = true; Save.NotePassive(10);
            }, Take = () => { Save.D.far = false; Save.NotePassive(10); } },
        new ShopItem { Id = 11, Name = "陷阱", Desc = "敌人掉下去一时会爬不出来", Rarity = "稀有", Icon = "it_trap", Active
            = true, Price = Economy.ItemPrices[11], Count = () => Save.D.trap, Give = () => { Save.D.trap++; Save.NoteActive(11); }
            , Take = () => { Save.D.trap = Mathf.Max(0, Save.D.trap - 1); Save.NoteActive(11); } },
        new ShopItem { Id = 12, Name = "十字坡灶", Desc = "占一格，慢慢出炊饼；天书能升灶", Rarity = "卓越", Icon = "it_st
            ove", Price = Economy.ItemPrices[12], Count = () => B(Save.D.stove), Give = () => { Save.D.stove = true; Save.NotePassi
            ve(12); }, Take = () => { Save.D.stove = false; Save.NotePassive(12); } },
        new ShopItem { Id = 13, Name = "招贤酒", Desc = "聚义时更常出好汉的字", Rarity = "史诗", Icon = "it_wine", Price = E
            conomy.ItemPrices[13], Count = () => B(Save.D.wine), Give = () => { Save.D.wine = true; Save.NotePassive(13); }, Take =
             () => { Save.D.wine = false; Save.NotePassive(13); } },
    };
}
/// <summary>山寨主界面：座次、章节、出征、好汉谱、十字坡酒店。</summary>
public sealed partial class HomeScreen : MonoBehaviour
{
    RectTransform _root;
    Text _coins, _stamina, _loadout;
    int _selected;
    Text _chapterName, _chapterInfo, _rankName, _stars;
    RectTransform _badge, _badgeHolder;
    int _badgeTier = -1;
    RectTransform _shop;
    Text _shopCoins, _shopStamina;
    Action<int> _onFight;
    Image _hudFace, _profileBar;
    Text _profileHint;
    RectTransform _avatarHit, _authHit;
    bool _homeArmed, _boardOwns;
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    float _armAt;
    Action _reloadBoard;
    public static HomeScreen Open(Transform host, Action<int> onFight)
    {
        var go = new GameObject("Home");
        go.transform.SetParent(host, false);
        var h = go.AddComponent<HomeScreen>();
        h._onFight = onFight;
        h.Build();
        return h;
    }
    void OnDestroy()
    {
        _reloadBoard = null;
        _authHit = null;
        WxSession.HideProfileButton();
        if (_root) Destroy(_root.gameObject);
    }
    static readonly Color Edge = Pal.Hex("2E2119");
    static readonly Color StarOn = Pal.Hex("FFD84A");
    static readonly Color StarOff = Pal.Hex("6E5D45");
    RectTransform Corner(float x, float y, string name)
    {
        var rt = Ui.Rect(_root, name);
        Ui.Anchor(rt, x, y);
        return rt;
    }
    void Build()
    {
        _root = Ui.Rect(Ui.Root, "Home");
        _root.anchorMin = Vector2.zero;
        _root.anchorMax = Vector2.one;
        _root.offsetMin = _root.offsetMax = Vector2.zero;
        Save.RefreshShopDay();
        PageBg(_root, 0f).SetAsFirstSibling();
        var tl = Corner(0f, 1f, "topLeft");
        BuildHud(tl, out _coins, out _stamina, true);
        MedalBtn(tl, "音", null, new Vector2(80, -226), 92, ToggleSound);
        MedalBtn(tl, "告", "公告", new Vector2(80, -372), 108, ShowNotice);
        _codexDot = MedalBtn(tl, "谱", "好汉谱", new Vector2(80, -534), 108, ShowCodex);
        MedalBtn(tl, "回", "回目", new Vector2(80, -696), 108, ShowChapters);
        _dailyDot = MedalBtn(tl, "战", "每日一战", new Vector2(80, -858), 108, ShowDailyBattle);
        if (!Save.DailyOpen) _dailyDot.transform.parent.GetComponent<Image>().color = new Color(0.62f, 0.6f, 0.58f);
        var tr = Corner(1f, 1f, "topRight");
        Ui.Medal(tr, 128, new Vector2(-88, -250), "shopDisc");
        var shopFace = Ui.Icon(tr, "merchant_bust", 150, new Vector2(-88, -236));
        Ui.Tap(shopFace, ShowShop);
        MedalLabel(tr, "神秘商人", new Vector2(-88, -332));
        BuildGm(tr);
        var top = Corner(0.5f, 1f, "title");
        var logo = Ui.Icon(top, "home_logo", 760, new Vector2(0, -322), "logo");
        Tween.Pop(logo.transform, 0.5f);
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        BuildRankPlate(top, new Vector2(0, -548));
        var hero = Corner(0.5f, 0.5f, "hero");
        hero.SetSiblingIndex(1);
        BuildHero(hero);
        var bottom = Corner(0.5f, 0f, "bottom");
        _loadout = Ui.Label(bottom, "", 30, Ui.GoldText, new Vector2(0, 556), new Vector2(940, 46));
        Ui.Stroke(_loadout, Pal.Hex("120B07"), 2f);
        BuildStart(bottom);
        SideBigIcon(Corner(0f, 0f, "rankEntry"), "榜", "排行榜", new Vector2(130, 200), ShowBoard);
        SideBigIcon(Corner(1f, 0f, "bagEntry"), "兵", "武器背包", new Vector2(-130, 200), ShowArmory);
        Refresh();
        LiveConfig.Refresh();
        LiveConfig.SubmitRank();
        if (!Save.Volatile && Save.RollDaily()) ShowDaily();
    }
    /// <summary>整页底图：梁山水墨，等比铺满到屏幕边，dim 是压在上面的暗度。</summary>
    static RectTransform PageBg(RectTransform parent, float dim)
    {
        var bg = Ui.Rect(parent, "bg");
        Ui.PinBleed(bg);
        var under = bg.gameObject.AddComponent<Image>();
        under.color = Pal.Hex("1A110B");
        under.raycastTarget = false;
        var art = Ui.Rect(bg, "art");
        var img = art.gameObject.AddComponent<Image>();
        img.sprite = Art.Img("home_bg");
        img.raycastTarget = false;
        var fit = art.gameObject.AddComponent<AspectRatioFitter>();
        fit.aspectMode = AspectRatioFitter.AspectMode.EnvelopeParent;
        fit.aspectRatio = Ui.DesignW / Ui.DesignH;
        if (dim > 0f)
        {
            var shade = Ui.Rect(bg, "dim");
            shade.anchorMin = Vector2.zero;
            shade.anchorMax = Vector2.one;
            shade.offsetMin = shade.offsetMax = Vector2.zero;
            var s = shade.gameObject.AddComponent<Image>();
            s.color = Pal.Hex("120B07").A(dim);
            s.raycastTarget = false;
        }
        return bg;
    }
    /// <summary>标题下的座次牌：左嵌勋章，中写座次，右排星；整块点开排行榜。</summary>
    void BuildRankPlate(RectTransform parent, Vector2 pos)
    {
        const float w = 600f, h = 84f;
        var plate = Ui.Rect(parent, "rankPlate");
        plate.anchoredPosition = pos;
        plate.sizeDelta = new Vector2(w, h);
        Ui.Box(plate, new Vector2(w + 8, h + 8), new Vector2(0, -7), Pal.Shadow.A(0.6f), null, "shadow");
        var face = Ui.Skin(plate, Ui.SkinSlot, new Vector2(w, h), Vector2.zero, Color.white, "face");
        face.raycastTarget = true;
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365         Ui.Tap(face, ShowBoard);
366         _rankName = Ui.Label(plate, "", 40, Ui.GoldText, new Vector2(-w / 2 + 92 + 130, 0), new Vector2(260, h), TextAnchor.Middl
367             eLeft, Fonts.Brush);
368         _rankName.resizeTextForBestFit = true;
369         _rankName.resizeTextMinSize = 28;
370         _rankName.resizeTextMaxSize = 40;
371         Ui.Stroke(_rankName, Edge, 2f);
372         _stars = Ui.Label(plate, "", 50, StarOn, new Vector2(w / 2 - 26 - 140, -2), new Vector2(280, h), TextAnchor.MiddleRight);
373         _stars.supportRichText = true;
374         Ui.Stroke(_stars, Edge, 2.5f);
375         _badgeHolder = Ui.Rect(plate, "badge");
376         _badgeHolder.anchoredPosition = new Vector2(-w / 2 + 18, 4);
377     }
378     const float HudY = -92f;
379     /// <summary>顶栏一行：头像、铜钱条、体力条（绿加号看广告补体力）。商人界面也复用。</summary>
380     void BuildHud(RectTransform tl, out Text coins, out Text stamina, bool avatar)
381     {
382         if (avatar)
383         {
384             var face = CirclePortrait(tl, WxSession.Avatar, new Vector2(80, HudY), 118f, out var hit);
385             if (tl.name == "topLeft")
386             {
387                 _hudFace = face;
388                 _avatarHit = hit;
389                 var mask = hit.Find("avatarMask");
390                 _profileBar = Ui.Box(mask, new Vector2(118, 36), new Vector2(0, -40), new Color(0.12f, 0.07f, 0.04f, 0.82f), null, "nameB
391             ar");
392                 var hint = Ui.Label(mask, "点我授权", 22, Pal.Hex("FFE7A8"), new Vector2(0, -40), new Vector2(110, 32));
393                 hint.resizeTextForBestFit = true;
394                 hint.resizeTextMinSize = 14;
395                 hint.resizeTextMaxSize = 22;
396                 hint.horizontalOverflow = HorizontalWrapMode.Overflow;
397                 hint.verticalOverflow = VerticalWrapMode.Truncate;
398                 Ui.Stroke(hint, Edge, 1.5f);
399                 _profileHint = hint;
400                 Ui.Tap(face, OnHomeAvatar);
401             }
402             else Ui.Tap(face, () => Toast(ProfileLine()));
403         }
404         float x = avatar ? 186 : 40;
405         HudPill(tl, x, 250, "coinBar");
406         Ui.Icon(tl, "coin", 92, new Vector2(x + 14, HudY));
407         coins = Ui.Label(tl, "", 44, Pal.Hex("FFE7A8"), new Vector2(x + 145, HudY), new Vector2(180, 76));
408         coins.fontStyle = FontStyle.Bold;
409         x += 280;
410         HudPill(tl, x, 330, "staBar");
411         Ui.Icon(tl, "ui_bolt", 72, new Vector2(x + 12, HudY));
412         stamina = Ui.Label(tl, "", 38, Pal.Hex("FFE7A8"), new Vector2(x + 152, HudY), new Vector2(230, 76));
413         stamina.fontStyle = FontStyle.Bold;
414         stamina.supportRichText = true;
415         var plus = Ui.Icon(tl, "ui_plus", 62, new Vector2(x + 316, HudY));
416         Ui.Tap(plus, () =>
```

```
        {
            if (Stamina.Current >= Stamina.Max) { Nope("体力已满"); return; }
            AdStub.Show("补充体力", () => { Stamina.Add(5); Refresh(); Toast("体力 +5"); });
        });
        AdStub.Badge(tl, new Vector2(x + 316, HudY - 50), "补充体力");
    }
    /// <summary>顶栏的深色圆角条，和座次牌同一套描边。left 是条的左端。</summary>
    static void HudPill(RectTransform parent, float left, float w, string name)
    {
        const float h = 72f;
        var c = new Vector2(left + w / 2, HudY);
        Ui.Skin(parent, Ui.SkinSlot, new Vector2(w, h), c, Color.white, name);
    }
    /// <summary>金边黑漆圆章入口：章心一个金字，下面挂一块小漆牌写全名。返回右上角的小红点，默认藏着。</summary>
    Image MedalBtn(RectTransform parent, string glyph, string label, Vector2 pos, float size, Action onClick)
    {
        var medal = Ui.Medal(parent, size, pos);
        Ui.Tap(medal, onClick);
        var g = Ui.Label(medal.transform, glyph, Mathf.RoundToInt(size * 0.5f), Ui.GoldText, new Vector2(0, 2), Vector2.one * size,
            TextAnchor.MiddleCenter, Fonts.Brush, "glyph");
        g.gameObject.AddComponent<Shadow>().effectColor = Color.black.A(0.6f);
        if (label != null) MedalLabel(medal.transform, label, new Vector2(0, -size * 0.5f - 4));
        var dot = Ui.Box(medal.transform, new Vector2(30, 30), new Vector2(size * 0.36f, size * 0.36f), Pal.Hex("E8392B"), Art.Circle,
            "redDot");
        dot.type = Image.Type.Simple;
        dot.gameObject.SetActive(false);
        return dot;
    }
    static void MedalLabel(Transform parent, string text, Vector2 pos, int font = 28)
    {
        if (text.Length > 3) font = Mathf.Min(font, 24);
        float w = text.Length * font + 26;
        Ui.Skin(parent, Ui.SkinSlot, new Vector2(w, font + 16), pos, Color.white, "labelPlate", 46f / ((font + 16) * 0.5f));
        Ui.Label(parent, text, font, Ui.GoldText, pos + new Vector2(0, 1), new Vector2(w, font + 16), TextAnchor.MiddleCenter, Font
            s.Brush, "labelText");
    }
    Image _codexDot, _dailyDot;
    void SideBigIcon(RectTransform parent, string glyph, string text, Vector2 pos, Action onClick)
    {
        var medal = Ui.Medal(parent, 150, pos);
        Ui.Tap(medal, onClick);
        var g = Ui.Label(medal.transform, glyph, 78, Ui.GoldText, new Vector2(0, 3), new Vector2(150, 150), TextAnchor.MiddleCent
            er, Fonts.Brush, "glyph");
        g.gameObject.AddComponent<Shadow>().effectColor = Color.black.A(0.6f);
        MedalLabel(medal.transform, text, new Vector2(0, -84), 32);
    }
    /// <summary>中央：梁山兄弟劫囚车救宋江的插画，站在空地上；下方左右切换回目。</summary>
    void BuildHero(RectTransform c)
    {
        c.anchoredPosition = new Vector2(0, 40);
        var glow = Ui.Box(c, new Vector2(1000, 600), new Vector2(0, 40), Pal.Hex("FFE9BE").A(0.28f), Art.SoftDot, "heroGlow");
```

```
        glow.type = Image.Type.Simple;
        var shadow = Ui.Box(c, new Vector2(760, 90), new Vector2(0, -190), new Color(0.1f, 0.06f, 0.03f, 0.45f), Art.SoftDot, "heroSha
            dow");
        shadow.type = Image.Type.Simple;
        var art = Ui.Icon(c, "home_hero", 860, new Vector2(10, 20), "heroArt");
        art.rectTransform.sizeDelta = new Vector2(860, 473);
        Bob(art.transform, 0f);
        _selected = Mathf.Clamp(Save.D.unlocked, 0, ChapterDef.StoryCount);
        var strip = Ui.Skin(c, Ui.SkinSlot, new Vector2(580, 96), new Vector2(0, -300), Color.white, "chapterStrip");
        _chapterName = Ui.Label(strip.transform, "", 46, Ui.GoldText, new Vector2(0, 2), new Vector2(460, 88), TextAnchor.MiddleC
            enter, Fonts.Brush);
        _chapterName.resizeTextForBestFit = true;
        _chapterName.resizeTextMinSize = 26;
        _chapterName.resizeTextMaxSize = 46;
        _chapterName.verticalOverflow = VerticalWrapMode.Truncate;
        Ui.Tap(strip, () => { if (_selected >= ChapterDef.StoryCount) ShowTeam(); else ShowChapters(); });
        _chapterInfo = Ui.Label(c, "", 29, Pal.Paper, new Vector2(0, -376), new Vector2(860, 44));
        Ui.Stroke(_chapterInfo, Pal.Hex("120B07"), 2f);
        ArrowBtn(c, "<", new Vector2(-350, -300), -1);
        ArrowBtn(c, ">", new Vector2(350, -300), 1);
    }
    void ArrowBtn(RectTransform parent, string glyph, Vector2 pos, int dir)
    {
        var b = Ui.Medal(parent, 92, pos, "arrow");
        Ui.Label(b.transform, glyph, 36, Ui.GoldText, new Vector2(dir * 3, 0), new Vector2(92, 92));
        Ui.Tap(b, () => SelectChapter(_selected + dir));
    }
    void SelectChapter(int i)
    {
        if (i < 0 || i > ChapterDef.StoryCount) return;
        if (i > Save.D.unlocked) { Nope(i >= ChapterDef.StoryCount ? "打完蓼儿洼，聚义厅才开张" : "先赢下上一回才能解锁");
            return; }
        _selected = i;
        RefreshChapter();
        Tween.Punch(_chapterName.transform, 0.15f);
    }
    void RefreshChapter()
    {
        if (_selected >= ChapterDef.StoryCount)
        {
            _chapterName.text = $"聚义厅 · 第{BattleView.ToCn(Save.D.endlessRound)}阵";
            var names = new List<string>();
            foreach (var id in Save.Team()) names.Add(HeroDef.Get(id).Name);
            _chapterInfo.text = $"阵容 {string.Join(" ", names)}　最好 {Save.D.endlessBest} 阵　点此换将";
            return;
        }
        var ch = ChapterDef.All[_selected];
        _chapterName.text = "第" + BattleView.ToCn(ch.Index + 1) + "回 · " + ch.Name;
        var nem = HeroDef.Get(ch.Boss.Nemesis).Name;
        bool fresh = ch.Joins.Length > 0 && !Save.Joined(ch.Joins[0]);
        _chapterInfo.text = fresh
            ? $"{ch.Act}　头目 {ch.Boss.Name}　克星 {nem}　胜后 {ch.Joins.Length} 人入伙"
```

```
            : $"{ch.Act}  头目 {ch.Boss.Name}  克星 {nem}  对手 {ch.RivalName}";
    }
    void BuildStart(RectTransform bottom)
    {
        var holder = Ui.Rect(bottom, "start");
        holder.anchoredPosition = new Vector2(0, 430);
        holder.sizeDelta = new Vector2(600, 150);
        var glow = Ui.Box(holder, new Vector2(860, 300), Vector2.zero, Pal.Hex("E8573F").A(0.32f), Art.SoftDot, "glow");
        glow.type = Image.Type.Simple;
        Ui.Box(holder, new Vector2(610, 156), new Vector2(0, -10), Pal.Shadow.A(0.7f), null, "shadow");
        var body = Ui.Rect(holder, "body");
        body.sizeDelta = holder.sizeDelta;
        Ui.Skin(body, Ui.SkinBtn, new Vector2(600, 150), Vector2.zero, Pal.Hex("C23A2A"), "face", 1.2f);
        Ui.Skin(body, Ui.SkinRim, new Vector2(600, 150), Vector2.zero, Color.white, "rim", 1.2f);
        foreach (float side in new[] { -1f, 1f })
        {
            var d = Ui.Box(body, new Vector2(30, 30), new Vector2(side * 252, 0), Ui.GoldText, Art.White, "diamond");
            d.rectTransform.localRotation = Quaternion.Euler(0, 0, 45);
            var dd = Ui.Box(body, new Vector2(12, 12), new Vector2(side * 252, 0), Pal.Hex("6B1A12"), Art.White, "diamondHole");
            dd.rectTransform.localRotation = Quaternion.Euler(0, 0, 45);
        }
        var label = Ui.Label(body, "开始游戏", 70, Pal.Paper, new Vector2(-28, 2), new Vector2(400, 120), TextAnchor.MiddleCenter,
            Fonts.Brush);
        Ui.Stroke(label, Pal.Hex("5A140C"), 3f);
        Ui.Icon(body, "ui_bolt", 56, new Vector2(168, 4));
        Ui.Stroke(Ui.Label(body, "-" + Stamina.Cost, 28, Ui.GoldText, new Vector2(196, -26), new Vector2(60, 40)), Pal.Hex("5A140C"
            ), 1.5f);
        var hit = Ui.Box(holder, new Vector2(600, 150), Vector2.zero, new Color(0, 0, 0, 0), null, "hit");
        Ui.Tap(hit, () => Fight(), body, 10f);
        Breathe(holder);
    }
    static void Bob(Transform t, float phase)
    {
        var basePos = ((RectTransform)t).anchoredPosition;
        Tween.Run(t, 1.6f, k => { if (t) ((RectTransform)t).anchoredPosition = basePos + new Vector2(0, Mathf.Sin((k + phase) * Ma
            thf.PI * 2) * 6f); },
            () => { if (t) { ((RectTransform)t).anchoredPosition = basePos; Bob(t, phase); } });
    }
    static void Breathe(Transform t)
    {
        Tween.Run(t, 1.4f, k => { if (t) t.localScale = Vector3.one * (1f + Mathf.Sin(k * Mathf.PI * 2) * 0.03f); }, () => { if (t) Breathe(
            t); });
    }
    void OnHomeAvatar()
    {
        if (WxSession.CanAskProfile && !LiveConfig.HasNick)
        {
            _homeArmed = false;
            WxSession.AllowPrivacyRetry();
            if (WxSession.PrivacyAgreed) ArmHome();
            else WxSession.ShowPrivacy(ok => { if (ok) ArmHome(); }, true);
            return;
```

```
        }
        Toast(ProfileLine());
    }
    void ArmHome()
    {
        if (_homeArmed || _boardOwns || !_avatarHit) return;
        if (!WxSession.CanAskProfile || !WxSession.PrivacyAgreed || LiveConfig.HasNick) return;
        if (WxSession.PrivacyShowing || AvatarBlocked()) return;
        _homeArmed = true;
        _armAt = Time.unscaledTime;
        WxSession.Cover(this, _avatarHit, OnProfile);
    }
    void SyncProfileButton()
    {
        if (_boardOwns) return;
        if (!WxSession.CanAskProfile || LiveConfig.HasNick)
        {
            if (_homeArmed)
            {
                _homeArmed = false;
                WxSession.HideProfileButton();
            }
            return;
        }
        if (WxSession.PrivacyShowing || AvatarBlocked())
        {
            if (_homeArmed)
            {
                _homeArmed = false;
                WxSession.HideProfileButton();
            }
            return;
        }
        if (_homeArmed)
        {
            if (!WxSession.ProfileButtonVisible && Time.unscaledTime - _armAt > 2f) _homeArmed = false;
            else return;
        }
        ArmHome();
    }
    bool AvatarBlocked()
    {
        if (!_root) return false;
        for (int i = 0; i < _root.childCount; i++)
        {
            var child = _root.GetChild(i);
            if (!child.gameObject.activeInHierarchy) continue;
            switch (child.name)
            {
                case "panel":
                case "confirm":
                case "shop":
```

```
            case "daily":
            case "armory":
            case "forge":
                return true;
        }
    }
    return false;
}
static string ProfileLine() => $"{LiveConfig.MyName} · {Save.RankName} · 胜 {Save.D.wins} / 战 {Save.D.battles}";
void ToggleSound()
{
    Sfx.On = !Sfx.On;
    Save.D.sfx = Sfx.On;
    Save.Write();
    if (Sfx.On) Sfx.Play(Snd.Tap);
    Toast(Sfx.On ? "声音已打开" : "声音已关闭");
}
const string DefaultNotice =
    "各位好汉：\n\n" +
    "《营救宋江》已更新，公告如下。\n\n" +
    "【更新内容】\n" +
    "1. 更新首页与战场画面。\n" +
    "2. 优化官兵行进时的朝向。\n" +
    "3. 修复排行榜名次显示错误。\n\n" +
    "【排行榜】\n" +
    "按战绩由高到低排列。\n" +
    "战绩相同，先达成者居前。\n" +
    "名单上的名次，与底部「我」一致。\n\n" +
    "【温馨提示】\n" +
    "· 登录微信后，同步昵称、头像和进度。\n" +
    "· 体力与每日次数，北京时间零点刷新。\n" +
    "· 若遇异常，退出后重新进入。\n\n" +
    "感谢支持。\n\n" +
    "《营救宋江》运营组\n" +
    "2026年11月7日";
void ShowNotice()
{
    string remote = LiveConfig.Notice;
    bool live = !string.IsNullOrEmpty(remote);
    Panel(live ? LiveConfig.NoticeTitle : "公告", out var card, new Vector2(900, 1300));
    // 内衬底边在 -624，正文停在它上面，避免最后一行压到木框。
    var body = Ui.Label(card, live ? remote.Replace("\\n", "\n") : DefaultNotice,
        34, Pal.Ink, new Vector2(0, -36), new Vector2(740, 1040), TextAnchor.UpperLeft);
    body.lineSpacing = 1.02f;
    body.verticalOverflow = VerticalWrapMode.Truncate;
}
void Fight() => Fight(_selected);
void Fight(int slot)
{
    if (!Stamina.Spend())
    {
        int left = AdStub.Left("补充体力");
```

```
            if (left == 0)
            {
                int sec = Stamina.SecondsToNext;
                Nope($"体力不够，今天的补体力广告也看完了。{sec / 60}分{sec % 60:00}秒后回 1 点");
                return;
            }
            string more = left > 0 ? $"\n今天还能补 {left} 次" : "";
            Confirm("体力不够了", "看段广告，补 5 点体力？ " + more, "看广告", () => AdStub.Show("补充体力", () => { Stamina.
            Add(5); Refresh(); }));
            return;
        }
        _onFight?.Invoke(slot);
    }
    float _refreshT;
    string _avatarSeen;
    void Update()
    {
        _refreshT -= Time.unscaledDeltaTime;
        if (_refreshT <= 0) { _refreshT = 1f; Refresh(); }
        SyncProfileButton();
    }
    void Refresh()
    {
        if (_codexDot) _codexDot.gameObject.SetActive(Save.CodexPending > 0);
        if (_dailyDot) _dailyDot.gameObject.SetActive(Save.DailyOpen && !Save.DailyWonToday);
        _coins.text = Save.D.coins.ToString();
        _stamina.text = StaminaText();
        if (_shopCoins != null) _shopCoins.text = _coins.text;
        if (_shopStamina != null) _shopStamina.text = _stamina.text;
        var next = Save.D.PeekLoadout();
        string carried = BattleScreen.ItemSummary(next);
        string worn = BattleScreen.GearSummary(next);
        if (worn.Length > 0) worn = "兵器架（" + worn + "）";
        string line = carried;
        if (worn.Length > 0) line = carried.Length > 0 ? worn + "　" + carried : worn;
        _loadout.text = line.Length > 0 ? "下一战随身：" + line : "";
        WxSession.EnsureAvatar();
        string avatar = WxSession.Avatar;
        if (_hudFace) Portraits.Set(_hudFace, avatar, true);
        if (_avatarSeen != null && avatar != _avatarSeen && avatar.Length > 0)
            _reloadBoard?.Invoke();
        _avatarSeen = avatar;
        bool showName = LiveConfig.HasNick || (WxSession.CanAskProfile && !LiveConfig.HasNick);
        if (_profileHint)
        {
            _profileHint.gameObject.SetActive(showName);
            if (LiveConfig.HasNick)
            {
                _profileHint.text = WxSession.Nick;
                WxSession.ApplyNickFont(_profileHint);
            }
            else _profileHint.text = "点我授权";
```

```
        }
        if (_profileBar) _profileBar.gameObject.SetActive(showName);
        _rankName.text = Save.RankName;
        _stars.text = StarRich(Save.D.stars);
        if (_badgeTier != Save.D.tier)
        {
            _badgeTier = Save.D.tier;
            if (_badge) Destroy(_badge.gameObject);
            _badge = Ui.RankBadge(_badgeHolder, Save.D.tier, Vector2.zero, 0.82f);
        }
        RefreshChapter();
    }
    static string StaminaText()
    {
        int s = Stamina.Current;
        if (s >= Stamina.Max) return $"{s} <size=30>/{Stamina.Max}</size>";
        int sec = Stamina.SecondsToNext;
        return $"{s} <size=30>/{Stamina.Max}  {sec / 60}:{sec % 60:00}</size>";
    }
    static string StarRich(int n) =>
        Ui.RankStars(Save.D.tier, n, ColorUtility.ToHtmlStringRGB(StarOn), ColorUtility.ToHtmlStringRGB(StarOff));
    // ———————————————————————————————— 浮层 ————————————————————————————————
    RectTransform Panel(string title, out RectTransform card, Vector2 size, Action onClose = null)
    {
        var layer = Ui.Fill(_root, new Color(0, 0, 0, 0.72f), true, "panel");
        card = Ui.WoodPanel(Ui.Fit(layer, size.y + 160), size, new Vector2(0, -20));
        Ui.Title(card, title, new Vector2(0, size.y / 2 - 10));
        var l = layer;
        Ui.CloseX(card, new Vector2(-size.x / 2 + 6, size.y / 2 - 4), () => { Destroy(l.gameObject); onClose?.Invoke(); });
        Tween.Pop(card, 0.25f);
        return layer;
    }
    static void Toast(string s) => Ui.Toast(s);
    static void Nope(string s) => Ui.Nope(s);
    void Confirm(string title, string body, string ok, Action onOk)
    {
        var layer = Ui.Fill(_root, new Color(0, 0, 0, 0.72f), true, "confirm");
        var card = Ui.WoodPanel(layer, new Vector2(840, 600), Vector2.zero);
        Ui.Title(card, title, new Vector2(0, 290), 80);
        Ui.Label(card, body, 42, Pal.Ink, new Vector2(0, 50), new Vector2(720, 200));
        Ui.CartoonBtn(card, ok, new Vector2(300, 110), new Vector2(-170, -170), Ui.BtnRed, () => { Destroy(layer.gameObject); onOk
            (); }, 46,
            ok.Contains("广告") ? "ui_video" : null);
        Ui.CartoonBtn(card, "算了", new Vector2(300, 110), new Vector2(170, -170), Ui.BtnDark, () => Destroy(layer.gameObject), 46
            );
        Tween.Pop(card, 0.25f);
    }
  }
}
// 文件: Assets/Scripts/Game/HomeDaily.cs
using System;
```

```
using System.Collections.Generic;
using Rescue.Sim;
using UnityEngine;
using UnityEngine.UI;
namespace Rescue
{
  /// <summary>每日一战：第五回之后开放。今天人人同一张图、同一个头目，首胜给铜钱和一件兵器，不动座次。</su
    mmary>
  public sealed partial class HomeScreen
  {
    static readonly Color CellPath = Pal.Hex("B98E5E");
    static readonly Color CellTile = Pal.Hex("F4EBD6");
    static readonly Color CellGrass = Pal.Hex("8DB563");
    static readonly Color CellWater = Pal.Hex("7FB5D6");
    static readonly Color CellRock = Pal.Hex("8E8A86");
    static readonly Color CellMarsh = Pal.Hex("6E9E86");
    public void ShowDailyBattle()
    {
      if (!Save.DailyOpen)
      {
        Nope($"赢下第{BattleView.ToCn(Save.DailyUnlock)}回才开放每日一战，还差 {Save.DailyUnlock - Save.D.unlocked}
        回");
        return;
      }
      var ch = ChapterDef.Daily(int.Parse(Save.Today));
      var layer = Panel("每日一战", out var card, new Vector2(920, 1420));
      var day = DateTime.ParseExact(Save.Today, "yyyyMMdd", null);
      Ui.Label(card, $"{day.Month}月{day.Day}日 · 今天人人同一阵", 34, Pal.InkSoft, new Vector2(0, 590), new Vector2(820, 50
        ));
      var name = Ui.Label(card, ch.Name.Replace("每日一战 · ", ""), 96, Pal.Ink, new Vector2(0, 500), new Vector2(820, 120), Text
        Anchor.MiddleCenter, Fonts.Brush);
      Ui.Stroke(name, Pal.Hex("FFF6E2"), 3f);
      MapPreview(card, ch, new Vector2(0, 270));
      string skills = ch.Boss.SkillText;
      Ui.Stroke(Ui.Label(card, $"头目 {ch.Boss.Title}{ch.Boss.Name}", 44, Pal.RedDeep, new Vector2(0, 60), new Vector2(820, 64),
        TextAnchor.MiddleCenter, Fonts.Brush), Pal.Hex("FFF6E2"), 2f);
      var sk = Ui.Label(card, skills, 28, Pal.Ink, new Vector2(0, -20), new Vector2(820, 90));
      sk.lineSpacing = 1.05f;
      var names = new List<string>();
      foreach (var id in ch.Pool) { var d = HeroDef.Get(id); if (!d.Hidden) names.Add(d.Name); }
      Ui.Label(card, $"克星 {HeroDef.Get(ch.Boss.Nemesis).Name}　本阵好汉 {string.Join(" ", names)}", 30, Pal.Gold, new Vect
        or2(0, -100), new Vector2(840, 50));
      bool won = Save.DailyWonToday;
      var box = Ui.Framed(card, new Vector2(800, 200), new Vector2(0, -250), won ? Pal.Hex("E2D2AE") : Pal.Hex("F6E2B0"), won
        ? Pal.Hex("8C6B4E") : Pal.Hex("D9822B"), 5f, "reward");
      var bt = box.transform;
      if (won)
      {
        Ui.Stroke(Ui.Label(bt, "今天已经赢过", 50, Pal.Willow, new Vector2(0, 36), new Vector2(760, 70), TextAnchor.MiddleCente
        r, Fonts.Brush), Pal.Hex("FFF6E2"), 2f);
        Ui.Label(bt, "再打只当练手，赢了给 15 铜钱，明天换新阵", 30, Pal.InkSoft, new Vector2(0, -36), new Vector2(760, 50));
      }
```

```
        else
        {
            Ui.Stroke(Ui.Label(bt, "今日首胜", 40, Pal.Hex("B5651D"), new Vector2(-250, 40), new Vector2(260, 60), TextAnchor.Middl
            eCenter, Fonts.Brush), Pal.Hex("FFF6E2"), 2f);
            Ui.Icon(bt, "coin", 84, new Vector2(-300, -36));
            Ui.Stroke(Ui.Label(bt, "×" + Save.DailyCoinsReward, 48, Pal.Hex("F2B53A"), new Vector2(-180, -36), new Vector2(160, 70
            ), TextAnchor.MiddleLeft), Edge, 2.5f).fontStyle = FontStyle.Bold;
            Ui.Icon(bt, Weapons.Icon(WeaponKind.Blade), 110, new Vector2(80, 0));
            Ui.Label(bt, "必掉一件\n入伙好汉的兵器", 30, Pal.Ink, new Vector2(250, 0), new Vector2(260, 100));
        }
        Ui.Label(card, "不动座次，输了也不掉星", 30, Pal.InkSoft, new Vector2(0, -385), new Vector2(800, 46));
        var go = Ui.CartoonBtn(card, "出 战", new Vector2(460, 132), new Vector2(-30, -520), Ui.BtnRed, () => Fight(ChapterDef.Da
            ilySlot), 64);
        Ui.Icon(go, "ui_bolt", 56, new Vector2(170, 0));
        Ui.Label(go, "-" + Stamina.Cost, 26, Pal.Hex("FFE7A8"), new Vector2(204, -26), new Vector2(60, 36));
        if (!won) Breathe(go);
    }
    /// <summary>今天这张地图的小样：路、白地、草地、水、石头一眼看清。</summary>
    static void MapPreview(RectTransform parent, ChapterDef ch, Vector2 center)
    {
        const float cell = 58f, gap = 4f;
        var board = new Board(ch.Map);
        float w = Board.Cols * (cell + gap) + gap, h = Board.Rows * (cell + gap) + gap;
        var frame = Ui.Framed(parent, new Vector2(w + 16, h + 16), center, Pal.Hex("5E4532"), Edge, 5f, "mapFrame");
        for (int c = 0; c < Board.Cols; c++)
            for (int r = 0; r < Board.Rows; r++)
            {
                var kind = board.Cells[c, r];
                Color col = kind switch
                {
                    Cell.Path => CellPath,
                    Cell.Tile => CellTile,
                    Cell.Water => CellWater,
                    Cell.Rock => CellRock,
                    Cell.Marsh => CellMarsh,
                    _ => CellGrass,
                };
                var pos = new Vector2(-w / 2 + gap + c * (cell + gap) + cell / 2, -h / 2 + gap + r * (cell + gap) + cell / 2);
                var img = Ui.Box(frame.transform, new Vector2(cell, cell), pos, col, null, "cell");
                img.pixelsPerUnitMultiplier = 2f;
                var at = new CellPos(c, r);
                if (at.Equals(board.Spawn)) Ui.Label(img.transform, "兵", 36, Pal.Paper, Vector2.zero, new Vector2(cell, cell), TextAnch
            or.MiddleCenter, Fonts.Brush);
                else if (at.Equals(board.Goal)) Ui.Label(img.transform, "宋", 40, Pal.RedDeep, Vector2.zero, new Vector2(cell, cell), Text
            Anchor.MiddleCenter, Fonts.Brush);
            }
    }
}
// 文件: Assets/Scripts/Game/HomeShop.cs
using System;
using System.Collections.Generic;
```

```
using Rescue.Sim;
using UnityEngine;
using UnityEngine.UI;
namespace Rescue
{
  /// <summary>神秘商人（孙二娘）：商店页三件货（最贵的一件看广告白拿），抽奖页九宫格转圈。</summary>
  public sealed partial class HomeScreen
  {
    static readonly string[] MerchantLines = { "诚信为本", "好货不等人", "客官里边请", "今日货今日用", "包子刚出笼", "童叟无欺" };
    const int ActiveSlots = 4;
    const int BagCols = 7;
    static readonly Color ShopWood = Pal.Hex("8C6B4E");
    static readonly Color ShopFace = Pal.Hex("C9B48C");
    static readonly Color LottoWood = Pal.Hex("8C6B4E");
    static readonly Color LottoFace = Pal.Hex("D8C49C");
    bool _lottoTab;
    Action _spinNow;
    /// <summary>验收截图用：跳过广告直接转一把。</summary>
    public void DebugSpin() => _spinNow?.Invoke();
    Text _say;
    RectTransform _bubble;
    /// <summary>结算后自动弹出，也能从主界面点孙二娘进来。</summary>
    public void ShowShop() => ShowShop(false);
    public void ShowShop(bool lotto)
    {
      Save.RefreshShopDay();
      _lottoTab = lotto;
      if (_shop) Destroy(_shop.gameObject);
      _shop = Ui.Fill(_root, Pal.Hex("120D0A").A(0.82f), true, "shop");
      var tl = Ui.Rect(_shop, "hud");
      Ui.Anchor(tl, 0f, 1f);
      BuildHud(tl, out _shopCoins, out _shopStamina, true);
      var body = Ui.Rect(Ui.Fit(_shop, 1740), "body");
      body.sizeDelta = new Vector2(960, 1500);
      body.anchoredPosition = new Vector2(0, -40);
      var bust = Ui.Icon(body, "merchant_bust", 230, new Vector2(30, 700));
      Bob(bust.transform, 0f);
      var bubble = Ui.Framed(body, new Vector2(230, 84), new Vector2(290, 760), Pal.Paper, Edge, 3f, "bubble");
      _bubble = (RectTransform)bubble.transform.parent;
      _say = Ui.Label(bubble.transform, MerchantLines[UnityEngine.Random.Range(0, MerchantLines.Length)], 32, Pal.Ink, Vector2.zero, new Vector2(220, 80));
      var panelHolder = Ui.Rect(body, "panelHolder");
      var content = Ui.Rect(body, "content");
      Ui.Title(body, "神秘商人", new Vector2(0, 590));
      Ui.CloseX(body, new Vector2(-445, 600), () => { if (_shop) Destroy(_shop.gameObject); _shop = null; _shopCoins = _shopStamina = null; });
      RectTransform panel = null;
      void Rebuild(string line)
      {
        if (panel) Destroy(panel.gameObject);
        panel = Ui.WoodPanel(panelHolder, new Vector2(940, 1110), new Vector2(0, 45), _lottoTab ? LottoWood : ShopWood, _lot
```

```
            toTab ? LottoFace : ShopFace);
            for (int i = content.childCount - 1; i >= 0; i--) Destroy(content.GetChild(i).gameObject);
            if (line != null) Say(line);
            BuildTabs(content, Rebuild);
            if (_lottoTab) BuildLotto(content, Rebuild);
            else BuildOffers(content, Rebuild);
            BuildBag(content, Rebuild);
            Refresh();
        }
        Rebuild(null);
        var tip = Ui.Label(_shop, "打完一局换新货。一场最多带两个主动、四个被动", 36, Pal.Paper, new Vector2(0, -640), new V
            ector2(940, 60));
        tip.fontStyle = FontStyle.Bold;
        Tween.Pop(body, 0.3f);
    }
    void Say(string line)
    {
        if (_say == null) return;
        _say.text = line;
        Tween.Punch(_bubble, 0.2f);
    }
    void BuildTabs(RectTransform content, Action<string> rebuild)
    {
        void Tab(string text, float x, bool on, bool lotto)
        {
            var face = Ui.Framed(content, new Vector2(170, 72), new Vector2(x, 500), on ? Ui.BtnYellow : Pal.Hex("A8865F"), Edge, 4f,
             "framedTab");
            Ui.Stroke(Ui.Label(face.transform, text, 40, on ? Edge : Pal.Paper, new Vector2(0, 2), new Vector2(170, 72), TextAnchor.Mid
            dleCenter, Fonts.Brush),
                on ? Pal.Hex("FFF2B8") : Pal.Hex("5E4532"), 1.5f);
            if (on) return;
            Ui.Tap(face, () => { _lottoTab = lotto; rebuild(lotto ? "手气旺不旺，转一把就知道" : "客官里边请"); });
        }
        Tab("商店", -100, !_lottoTab, false);
        Tab("抽奖", 100, _lottoTab, true);
        if (Save.LottoLeft > 0 && !_lottoTab)
        {
            var dot = Ui.Box(content, new Vector2(30, 30), new Vector2(178, 528), Pal.Hex("E8392B"), Art.Circle, "redDot");
            dot.type = Image.Type.Simple;
        }
    }
    /// <summary>今天三件货里最贵的那件，可以看广告白拿。</summary>
    static int FreeOfferId()
    {
        int best = -1, price = -1;
        foreach (var id in Save.D.shopOffer)
            if (ShopItem.All[id].Price > price) { price = ShopItem.All[id].Price; best = id; }
        return best;
    }
    void BuildOffers(RectTransform content, Action<string> rebuild)
    {
        int free = FreeOfferId();
```

```
        for (int i = 0; i < Save.D.shopOffer.Count; i++)
        {
            var it = ShopItem.All[Save.D.shopOffer[i]];
            float y = 360 - i * 190;
            var card = Ui.Framed(content, new Vector2(820, 172), new Vector2(0, y), Pal.Hex("F2E7CA"), Pal.Hex("5A4632"), 5f, "offer
            ");
            var t = card.transform;
            Ui.Box(t, new Vector2(148, 148), new Vector2(-322, 0), Edge, Art.Circle, "discEdge").type = Image.Type.Simple;
            Ui.Box(t, new Vector2(138, 138), new Vector2(-322, 0), it.DiscColor, Art.Circle, "disc").type = Image.Type.Simple;
            Ui.Icon(t, it.Icon, 104, new Vector2(-322, 0));
            Ui.Stroke(Ui.Label(t, it.Rarity, 38, it.RarityColor, new Vector2(-350, 62), new Vector2(120, 50), TextAnchor.MiddleCenter, F
            onts.Brush), Edge, 2.5f);
            var name = Ui.Label(t, it.Name, 40, Pal.Ink, new Vector2(-60, 44), new Vector2(300, 56), TextAnchor.MiddleLeft);
            name.fontStyle = FontStyle.Bold;
            Ui.Label(t, (it.Active ? "主动:" : "被动:") + it.Desc, 28, Pal.Hex("4A3A2C"), new Vector2(-50, -24), new Vector2(320, 90), Te
            xtAnchor.UpperLeft)
                .lineSpacing = 1.05f;
            bool sold = Save.D.shopSold.Contains(it.Id) || Save.D.shopFree.Contains(it.Id);
            if (sold)
            {
                var gone = Ui.Framed(t, new Vector2(212, 104), new Vector2(276, 0), Pal.Hex("4D4642"), Edge, 5f, "framedSold");
                Ui.Label(gone.transform, "已售完", 44, Pal.Paper, Vector2.zero, new Vector2(200, 100)).fontStyle = FontStyle.Bold;
                continue;
            }
            if (it.Id == free)
            {
                var adBtn = Ui.CartoonBtn(t, "免费\n领取", new Vector2(212, 108), new Vector2(276, 0), Ui.BtnOrange, () =>
                {
                    if (!CanTake(it)) return;
                    WarnPassiveFull(it, () => AdStub.Show("免费领取" + it.Name, () =>
                    {
                        it.Give();
                        Save.D.shopFree.Add(it.Id);
                        Save.Write();
                        Analytics.ShopBuy(ItemKey(it), "ad", 0);
                        Sfx.Play(Snd.Merge);
                        Toast(BoughtLine(it));
                        rebuild("白送的，客官拿好！");
                    }));
                }, 40, "ui_video");
                adBtn.Find("body/text").GetComponent<Text>().lineSpacing = 0.9f;
                AdStub.Badge(adBtn, new Vector2(80, 58), "免费领取");
                Shine(adBtn);
                continue;
            }
            var btn = Ui.Framed(t, new Vector2(212, 104), new Vector2(276, 0), Ui.BtnYellow, Edge, 5f, "framedBuy");
            Ui.Box(btn.transform, new Vector2(196, 14), new Vector2(0, 34), Color.white.A(0.35f), null, "gloss");
            Ui.Icon(btn.transform, "coin", 70, new Vector2(-52, 0));
            var price = Ui.Label(btn.transform, it.Price.ToString(), 46, Edge, new Vector2(34, 0), new Vector2(120, 100));
            price.fontStyle = FontStyle.Bold;
            if (Save.D.coins < it.Price) price.color = Pal.Hex("A33A2A");
            Ui.Tap(btn, () =>
```

```
            {
                if (Save.D.coins < it.Price) { Nope("铜钱不够，打一局再来"); return; }
                if (!CanTake(it)) return;
                WarnPassiveFull(it, () =>
                {
                    if (Save.D.coins < it.Price) return;
                    Save.D.coins -= it.Price;
                    it.Give();
                    Save.D.shopSold.Add(it.Id);
                    Save.Write();
                    Analytics.ShopBuy(ItemKey(it), "coin", it.Price);
                    Sfx.Play(Snd.Merge);
                    Toast(BoughtLine(it));
                    rebuild("客官好眼力！");
                });
            });
        }
    }
    static bool CanTake(ShopItem it)
    {
        if (!it.Active && it.Owned()) { Nope("这件已经有了"); return false; }
        if (it.Active && it.Count() == 0 && ActiveKinds() >= ActiveSlots) { Nope("背包里的主动道具最多四种"); return false; }
        if (!it.Active && it.Count() == 0 && Save.PassiveKinds >= Save.MaxPassiveBag) { Nope("背包里的被动道具最多七种"); re
            turn false; }
        return true;
    }
    static string ItemKey(ShopItem it) => it.Icon + ":" + it.Id;
    /// <summary>被动上场已满四个时，买之前先说清楚：买下只进背包，要带上得换下一个。</summary>
    void WarnPassiveFull(ShopItem it, Action go)
    {
        if (it.Active || Save.PassiveFieldCount < Save.MaxPassiveField) { go(); return; }
        Confirm("被动已带满四个", $"「{it.Name}」买下先放背包，\n这场不生效。要带它，\n在背包里点掉一个再点它", "
            照样要", go);
    }
    /// <summary>广告按钮上一道斜光来回扫，提醒「这个免费」。</summary>
    static void Shine(RectTransform btn)
    {
        var star = Ui.Icon(btn, "ui_sparkle", 54, new Vector2(btn.sizeDelta.x * 0.5f - 6, btn.sizeDelta.y * 0.5f - 6), "sparkle");
        Tween.Run(star, 1.2f, k => { if (star) { star.transform.localScale = Vector3.one * (0.6f + Mathf.Sin(k * Mathf.PI) * 0.5f); star.t
            ransform.localRotation = Quaternion.Euler(0, 0, k * 90f); } },
            () => { if (star) { Destroy(star.gameObject); Shine(btn); } });
    }
    static int ActiveKinds()
    {
        int n = 0;
        foreach (var it in ShopItem.All) if (it.Active && it.Count() > 0) n++;
        return n;
    }
    // ———————————————————————————————— 抽奖 ————————————————————————————————
    static readonly Vector2Int[] Ring =
    {
```

```
        new Vector2Int(0, 0), new Vector2Int(1, 0), new Vector2Int(2, 0), new Vector2Int(2, 1),
        new Vector2Int(2, 2), new Vector2Int(1, 2), new Vector2Int(0, 2), new Vector2Int(0, 1),
    };
    static Vector2 LottoPos(Vector2Int g) => new Vector2(-198 + g.x * 198, 368 - g.y * 198);
    void BuildLotto(RectTransform content, Action<string> rebuild)
    {
        var pool = Save.LottoPool();
        Ui.Framed(content, new Vector2(620, 620), new Vector2(0, 170), Pal.Hex("2A1C13"), Ui.GoldLine, 5f, "lottoBg");
        var cells = new RectTransform[pool.Count];
        for (int k = 0; k < pool.Count && k < Ring.Length; k++)
        {
            var it = ShopItem.All[pool[k]];
            var face = Ui.Framed(content, new Vector2(176, 176), LottoPos(Ring[k]), Color.Lerp(it.RarityColor, Pal.Hex("2A1C13"), 0.2
            2f), Edge, 5f, "framedLotto");
            face.pixelsPerUnitMultiplier = 1.2f;
            Ui.Box(face.transform, new Vector2(160, 60), new Vector2(0, -50), Color.black.A(0.12f), null, "band").pixelsPerUnitMultipli
            er = 1.2f;
            Ui.Icon(face.transform, it.Icon, 112, new Vector2(0, 16));
            var nm = Ui.Label(face.transform, it.Name, it.Name.Length > 3 ? 26 : 32, Pal.Paper, new Vector2(0, -58), new Vector2(176,
            44), TextAnchor.MiddleCenter, Fonts.Brush);
            Ui.Stroke(nm, Edge, 2f);
            Ui.Tap(face, () => Toast(it.Name + ":  " + it.Desc));
            cells[k] = (RectTransform)face.transform.parent;
        }
        var hl = Ui.Box(content, new Vector2(204, 204), LottoPos(Ring[0]), Pal.Hex("FFF27A"), Art.Frame, "highlight");
        hl.pixelsPerUnitMultiplier = 0.5f;
        hl.gameObject.SetActive(false);
        int left = Save.LottoLeft;
        var center = Ui.Framed(content, new Vector2(176, 176), LottoPos(new Vector2Int(1, 1)), left > 0 ? Ui.BtnRed : Pal.Hex("8E8A
            86"), Edge, 5f, "framedSpin");
        center.pixelsPerUnitMultiplier = 1.2f;
        Ui.Stroke(Ui.Label(center.transform, "抽奖", 52, Pal.Paper, new Vector2(0, 34), new Vector2(176, 70), TextAnchor.MiddleCent
            er, Fonts.Brush), Edge, 3f);
        Ui.Icon(center.transform, "ui_video", 74, new Vector2(0, -26));
        Ui.Label(center.transform, left > 0 ? $"今日剩{left}次" : "明日再来", 24, Pal.Paper, new Vector2(0, -70), new Vector2(176, 30
            )).fontStyle = FontStyle.Bold;
        if (left > 0) Breathe(center.transform.parent);
        _spinNow = () => Spin(content, cells, hl, pool, rebuild);
        Ui.Tap(center, () =>
        {
            if (Save.LottoLeft <= 0) { Nope("今天的抽奖次数用完了，明天再来"); return; }
            AdStub.Show("抽奖", () => Spin(content, cells, hl, pool, rebuild));
        });
        Ui.Label(content, "我的道具:", 34, Pal.Ink, new Vector2(-262, -186), new Vector2(300, 50), TextAnchor.MiddleLeft).fontStyle
            = FontStyle.Bold;
        for (int i = 0; i < ShopItem.Rarities.Length; i++)
            Ui.Tag(content, ShopItem.Rarities[i], ShopItem.ColorOf(ShopItem.Rarities[i]), new Vector2(-50 + i * 116, -186), 96, 26);
    }
    void Spin(RectTransform content, RectTransform[] cells, Image hl, List<int> pool, Action<string> rebuild)
    {
        int total = 0;
        foreach (var id in pool) total += ShopItem.All[id].Weight;
```

```
        int roll = UnityEngine.Random.Range(0, total), target = 0;
        for (int k = 0; k < pool.Count; k++)
        {
            roll -= ShopItem.All[pool[k]].Weight;
            if (roll < 0) { target = k; break; }
        }
        Save.D.lottoUsed++;
        Save.Write();
        var block = Ui.Fill(_shop, Color.clear, true, "spinBlock");
        hl.gameObject.SetActive(true);
        int steps = 8 * 3 + target;
        int last = -1;
        Say("转起来咯！");
        Tween.Run(hl, 2.8f, k =>
        {
            int idx = Mathf.Min(steps, Mathf.FloorToInt(Tween.OutCubic(k) * (steps + 0.999f)));
            if (idx == last) return;
            last = idx;
            hl.rectTransform.anchoredPosition = LottoPos(Ring[idx % 8]);
            Sfx.Play(Snd.Tap, 0.35f);
        }, () =>
        {
            var it = ShopItem.All[pool[target]];
            var cell = cells[target];
            Tween.Punch(cell, 0.4f, 0.25f);
            var rays = Ui.Icon(content, "ui_rays", 330, cell.anchoredPosition, "rays");
            rays.color = Color.white.A(0.85f);
            rays.transform.SetSiblingIndex(cell.GetSiblingIndex());
            Ui.Spin(rays.transform, 90f);
            Sfx.Play(Snd.Rain);
            string got = GiveLotto(it);
            Toast(got);
            Tween.Run(block, 1.3f, k => { }, () =>
            {
                if (block) Destroy(block.gameObject);
                rebuild(it.Rarity == "传说" || it.Rarity == "史诗" ? "哎哟，手气真旺！" : "恭喜客官！");
            });
        });
    }
    /// <summary>抽中的发进背包；主动种类满了、被动已有或上场已带满四个，就折成铜钱。</summary>
    static string GiveLotto(ShopItem it)
    {
        bool full = it.Active && it.Count() == 0 && ActiveKinds() >= ActiveSlots;
        bool dup = !it.Active && it.Owned();
        bool fifth = !it.Active && !dup && (Save.PassiveFieldCount >= Save.MaxPassiveField || Save.PassiveKinds >= Save.MaxPa
            ssiveBag);
        if (full || dup || fifth)
        {
            int coins = Economy.LottoRefund(it.Price);
            Save.D.coins += coins;
            Save.Write();
            Analytics.ShopBuy(ItemKey(it), "lotto_coin", coins);
```

```
            string why = full ? "主动道具已满" : dup ? "已经有了" : "被动已带满四个";
            return $"抽中「{it.Name}」，{why}，折铜钱 {coins}";
        }
        it.Give();
        Save.Write();
        Analytics.ShopBuy(ItemKey(it), "lotto", 0);
        if (it.Active && !Save.ActiveFielded(it.Id)) return $"抽中「{it.Rarity} · {it.Name}」，上场已满，先放背包";
        return $"抽中「{it.Rarity} · {it.Name}」！";
    }
    static string BoughtLine(ShopItem it)
    {
        if (it.Active && !Save.ActiveFielded(it.Id)) return it.Name + "已放进背包，上场已满两个";
        if (!it.Active && !Save.PassiveFielded(it.Id)) return it.Name + "已放进背包，被动上场已满四个";
        return "成功获得" + it.Name;
    }
    /// <summary>我的道具：主动一行（上场两件，背包四种）、被动一行（上场四件，背包七种），点图标决定带不带；今天买的可以点红减号退货。</summary>
    void BuildBag(RectTransform content, Action<string> rebuild)
    {
        if (!_lottoTab)
        {
            Ui.Label(content, "我的道具 · 点图标决定上场", 32, Pal.Ink, new Vector2(-190, -186), new Vector2(480, 50), TextAnchor.MiddleLeft).fontStyle = FontStyle.Bold;
            var count = Ui.Label(content, $"主动 {Save.D.activeOn.Count}/{Save.MaxActiveField}　被动 {Save.PassiveFieldCount}/{Save.MaxPassiveField}", 28, Pal.Hex("4A3A2C"),
                new Vector2(250, -186), new Vector2(320, 50), TextAnchor.MiddleRight);
            count.fontStyle = FontStyle.Bold;
        }
        Ui.Framed(content, new Vector2(790, 248), new Vector2(26, -330), Pal.Hex("3A2A20"), Pal.Hex("5A4632"), 5f, "bag");
        var side = _lottoTab ? Pal.Hex("3A2A20") : Pal.Hex("3A2A20");
        Ui.Label(content, "主\n动", 30, side, new Vector2(-400, -270), new Vector2(44, 90)).fontStyle = FontStyle.Bold;
        Ui.Label(content, "被\n动", 30, side, new Vector2(-400, -390), new Vector2(44, 90)).fontStyle = FontStyle.Bold;
        for (int row = 0; row < 2; row++)
        {
            var items = new List<ShopItem>();
            foreach (var it in ShopItem.All) if (it.Active == (row == 0) && it.Count() > 0) items.Add(it);
            for (int c = 0; c < BagCols; c++)
            {
                var pos = new Vector2(-304 + c * 110, row == 0 ? -270 : -390);
                Ui.Box(content, new Vector2(102, 102), pos, Pal.Hex("53392A"), null, "cell").pixelsPerUnitMultiplier = 1.4f;
                if (c >= items.Count) continue;
                var it = items[c];
                bool fielded = it.Active ? Save.ActiveFielded(it.Id) : Save.PassiveFielded(it.Id);
                var cell = Ui.Box(content, new Vector2(96, 96), pos, fielded ? it.DiscColor : it.DiscColor.A(0.4f), null, "item");
                cell.pixelsPerUnitMultiplier = 1.4f;
                var icon = Ui.Icon(cell.transform, it.Icon, 80, Vector2.zero);
                if (!fielded) icon.color = new Color(1f, 1f, 1f, 0.45f);
                int n = it.Count();
                if (n > 1) Ui.Stroke(Ui.Label(cell.transform, "×" + n, 26, Pal.Paper, new Vector2(22, -30), new Vector2(60, 34)), Edge, 2f);
                if (fielded)
                {
```

```
                var tag = Ui.Box(cell.transform, new Vector2(40, 28), new Vector2(-28, 34), Pal.Hex("C0392B"), null, "on");
                tag.pixelsPerUnitMultiplier = 1.4f;
                Ui.Label(tag.transform, "上", 22, Pal.Paper, Vector2.zero, new Vector2(40, 28));
            }
            Ui.Tap(cell, () =>
            {
                bool was = it.Active ? Save.ActiveFielded(it.Id) : Save.PassiveFielded(it.Id);
                string err;
                bool ok = it.Active ? Save.ToggleActiveField(it.Id, out err) : Save.TogglePassiveField(it.Id, out err);
                if (!ok) { Nope(err); return; }
                if (!was) Toast(it.Name + ":  " + it.Desc);
                rebuild(was ? it.Name + "这场不带" : it.Name + "这场带上");
            });
            if (!Save.D.shopSold.Contains(it.Id)) continue;
            var minus = Ui.Box(cell.transform, new Vector2(42, 42), new Vector2(42, 42), Pal.Hex("D7362B"), Art.Circle, "minus");
            minus.type = Image.Type.Simple;
            Ui.Box(minus.transform, new Vector2(22, 7), Vector2.zero, Color.white, Art.White, "bar").type = Image.Type.Simple;
            Ui.Tap(minus, () =>
            {
                it.Take();
                Save.D.shopSold.Remove(it.Id);
                Save.D.coins += it.Price;
                Save.Write();
                Analytics.ShopBuy(ItemKey(it), "refund", -it.Price);
                Toast($"已退回{it.Name}，铜钱 +{it.Price}");
                rebuild("下回再来哟");
            });
          }
        }
        var plank = Ui.Icon(content, "ui_plank", 360, new Vector2(-304 + 5 * 110, -270), "plank");
        plank.rectTransform.sizeDelta = new Vector2(350, 120);
        plank.preserveAspect = false;
        plank.rectTransform.localRotation = Quaternion.Euler(0, 0, -2);
        Ui.Tap(plank, () => Toast("主动背包最多四种，一场带两个；被动一场带四个"));
      }
    }
}
// 文件: Assets/Scripts/Sim/Defs.cs
using System;
using System.Collections.Generic;
namespace Rescue.Sim
{
    public enum UnitKind { Dao, Qiang, Gong, Qi }
    /// <summary>Shovel 是鲁智深的方便铲：拖到紧挨白地的草地上开出新地。Stove 是十字坡灶，占一格慢慢出炊饼。</s
        ummary>
    public enum PieceType { Unit, HeroChar, Shovel, Stove }
    public sealed class Piece
    {
        static int _nextId = 1;
        public readonly int Id;
        public PieceType Type;
        public UnitKind Kind;
```

```
    public char Char;
    public int Level = 1;
    /// <summary>被头目定身，剩余秒数。期间不出手，仍可拖动。</summary>
    public float StunT;
    public Piece() { Id = _nextId++; }
    public static Piece Unit(UnitKind k, int level = 1) => new Piece { Type = PieceType.Unit, Kind = k, Level = level };
    public static Piece Hero(char c, int level = 1) => new Piece { Type = PieceType.HeroChar, Char = c, Level = level };
    public static Piece Shovel() => new Piece { Type = PieceType.Shovel };
    public static Piece Stove() => new Piece { Type = PieceType.Stove };
    public char Glyph => Type switch
    {
        PieceType.Unit => UnitDef.Get(Kind).Glyph,
        PieceType.HeroChar => Char,
        PieceType.Stove => '灶',
        _ => '铲',
    };
}
public sealed class UnitDef
{
    public UnitKind Kind;
    public char Glyph;
    public string Role;
    public float Range, Cooldown, Damage;
    public int Pierce = 1;
    public float Splash;
    public float Knockback;
    public const int MaxLevel = 5;
    public float DamageAt(int level) => Damage * (float)Math.Pow(Tuning.LevelMul, level - 1);
    public float CooldownAt(int level) => Tuning.LevelCd(Cooldown, level);
    public string TypeLabel => Pierce > 1 ? "穿透" : Splash > 0 ? "范围" : Knockback > 0 ? "击退" : "单体";
    /// <summary>御敌千里只放大超过两格的远程。骑外沿正好两格，仍按近战，不翻倍。</summary>
    public float RangeWith(bool far) => far && Range > 2f ? Range * 2f : Range;
    static readonly UnitDef[] All =
    {
        // 圈从本格中心量。本格边在 0.5：刀再向外盖满一格，外沿 1.5；骑再向外一格半，外沿 2；弓够到三格外的格心。
        new UnitDef { Kind = UnitKind.Dao, Glyph = '刀', Role = "近战劈砍，附带小范围溅射", Range = 1.5f, Cooldown = 0.85f,
            Damage = 3.15f, Splash = 0.5f },
        new UnitDef { Kind = UnitKind.Qiang, Glyph = '枪', Role = "直线突刺，穿透三人", Range = 2.15f, Cooldown = 1.05f, Da
            mage = 2.7f, Pierce = 3 },
        new UnitDef { Kind = UnitKind.Gong, Glyph = '弓', Role = "远程放箭，射程三格", Range = 3.2f, Cooldown = 1.22f, Dam
            age = 2.7f },
        new UnitDef { Kind = UnitKind.Qi, Glyph = '骑', Role = "快速冲撞，击退敌人", Range = 2f, Cooldown = 0.42f, Damage =
            1.12f, Knockback = 0.22f },
    };
    public static UnitDef Get(UnitKind k) => All[(int)k];
}
public enum EnemyKind { Zu, Chai, Jia, Bu, Boss }
public sealed class EnemyDef
{
    public EnemyKind Kind;
    public string Glyph;
    public float HpMul, Speed, Armor;
```

```
    public int LeakDamage, Reward;
    static readonly EnemyDef[] All =
    {
        new EnemyDef { Kind = EnemyKind.Zu, Glyph = "卒", HpMul = 1f, Speed = 1.0f, LeakDamage = 1, Reward = 1 },
        new EnemyDef { Kind = EnemyKind.Chai, Glyph = "差", HpMul = 0.6f, Speed = 1.55f, LeakDamage = 1, Reward = 1 },
        new EnemyDef { Kind = EnemyKind.Jia, Glyph = "甲", HpMul = 2.3f, Speed = 0.74f, Armor = 0.35f, LeakDamage = 1, Rew
            ard = 2 },
        new EnemyDef { Kind = EnemyKind.Bu, Glyph = "捕", HpMul = 9f, Speed = 0.85f, LeakDamage = 2, Reward = 5 },
        new EnemyDef { Kind = EnemyKind.Boss, Glyph = "", HpMul = 9.5f, Speed = 0.45f, LeakDamage = 2, Reward = 20 },
    };
    public static EnemyDef Get(EnemyKind k) => All[(int)k];
}
public static class Tuning
{
    /// <summary>每边 8 列 5 行，和原作一样紧，摆错一格就疼。</summary>
    public const int Cols = 8;
    public const int Rows = 5;
    public const int CampSlots = 5;
    /// <summary>一颗心就是 1 点血。开局三颗，漏一只小怪掉一颗。</summary>
    public const int MaxHp = 3;
    public const int HeartHp = 1;
    /// <summary>续命丹加的心。3 + 8 = 11，第一波十个卒全放过去还剩 1 颗。</summary>
    public const int XumingHearts = 8;
    /// <summary>原作开局 20 包子、首次征兵 10：摆完第一轮手里还剩 10，第一波打完就能接上第二次。</summary>
    public const int StartBuns = 20;
    public const int StartCost = 10;
    /// <summary>杀敌入账倍率：一局大约聚义六到八次，每次都要想清楚再按。</summary>
    public const float KillGold = 0.8f;
    public const int CostStep = 3;
    /// <summary>掉一颗心补 6 个炊饼。只够喘口气，不够靠漏怪回本。</summary>
    public const int RainBuns = 6;
    public const float HeroCharChance = 0.24f;
    /// <summary>带着招贤酒时好汉字的几率。保底照旧，小兵仍占大半。</summary>
    public const float WineCharChance = 0.32f;
    /// <summary>十字坡灶：一级到三级的出饼间隔、每次几个、这一局最多出多少。</summary>
    public static readonly float[] StoveGap = { 10f, 8.5f, 7f };
    public static readonly int[] StoveYield = { 1, 1, 2 };
    public static readonly int[] StoveCap = { 12, 14, 16 };
    public const int StoveMaxLevel = 3;
    /// <summary>聚义自带铲子很少，白地主要靠看广告开。看完一次固定给两把。</summary>
    public const float ShovelChance = 0.05f;
    public const int AdShovels = 2;
    public const int GoldPity = 5;
    public const float PairBias = 0.4f;
    public const int MaxWaveCount = 24;
    /// <summary>每级伤害。比攻速陡，但不再一合成就翻两倍多，免得满级把血量曲线盖过去。</summary>
    public const double LevelMul = 1.88;
    /// <summary>每级攻速。连乘，五级大约比一级快七成半，出手节奏听得出来。</summary>
    public const float SpeedMul = 1.15f;
    public const float BossNemesisMul = 5f;
    /// <summary>出兵口草堆盖住的那段路。怪走出之前看不见，也不能被打到。</summary>
    public const float SpawnCover = 0.72f;
```

```
    /// <summary>开局先让探马沿路走一圈，玩家趁这段时间征兵布阵，再出第一波。</summary>
    public const float FirstWaveAt = 12f;
    public const float ScoutStart = 0.8f;
    public const float ScoutWalk = 7.5f;
    /// <summary>主动道具放完要转完这一圈才能再放。攻速符和它的十二秒效果一样长，刚好接上。</summary>
    public const float CdMeng = 8f;
    public const float CdBaozi = 6f;
    public const float CdShu = 10f;
    public const float CdFu = 12f;
    public const float CdTrap = 8f;
    /// <summary>前两波之后留口气看局面，后面逐步收紧。</summary>
    public static float WaveGap(int wave) => wave <= 2 ? 8f : wave <= 5 ? 7f : 6f;
    /// <summary>同一波里两个官兵的间隔：头几波稀疏，越往后越密。</summary>
    public static float SpawnInterval(int wave) => Math.Max(0.28f, wave <= 2 ? 1.3f - 0.1f * wave : 0.95f - 0.03f * wave);
    /// <summary>等级换算后的出手间隔。一级用基础值，之后每级按 SpeedMul 加快。</summary>
    public static float LevelCd(float cooldown, int level)
    {
        if (level < 1) level = 1;
        if (level > UnitDef.MaxLevel) level = UnitDef.MaxLevel;
        return cooldown / (float)Math.Pow(SpeedMul, level - 1);
    }
    /// <summary>普通官兵的基础血。前四波缓一缓，之后每波另加一截，第八波是开局的三倍半，一波比一波难。</su
        mmary>
    public static float WaveHp(int wave)
    {
        if (wave < 1) wave = 1;
        float grow = (float)Math.Pow(1.15, wave - 1);
        float extra = 1f + 0.075f * Math.Max(0, wave - 4);
        return 8.4f * grow * extra;
    }
}
/// <summary>
/// 梁山座次：十一个大阶各分叁、贰、壹三小阶，顶上是不封顶的「天魁星」。
/// 每一小阶都是五星，集满五星升下一小阶；败了掉一星，没星再败降一小阶，但不会跌出当前大阶。
/// </summary>
public static class RankRules
{
    public static readonly string[] Majors =
        { "喽啰", "小校", "都头", "提辖", "步军头领", "水军头领", "马军头领", "八骠骑", "五虎将", "地煞星", "天罡星", "天魁星"
            };
    public static readonly string[] MajorTips =
    {
        "山寨里跑腿的小兵", "带十来个喽啰的小头目", "县里的都头，如武松、朱仝、雷横",
        "军中提辖，如鲁达、杨志", "梁山步军头领，如鲁智深、武松", "梁山水军头领，如李俊、阮氏三雄",
        "梁山马军头领，如史进、穆弘", "马军八骠骑兼先锋使", "马军五虎上将",
        "七十二地煞之列", "三十六天罡之列", "天魁星呼保义，梁山之主",
    };
    static readonly string[] Subs = { "叁", "贰", "壹" };
    public const int SubPerMajor = 3;
    public const int TopMajor = 11;
    /// <summary>天魁星的扁平序号，也是最后一阶。</summary>
    public const int Top = TopMajor * SubPerMajor;
```

```
    public const int Count = Top + 1;
    /// <summary>界面上星条最多画几颗；天魁星的星数改成数字。</summary>
    public const int MaxStars = 5;
    public static int Major(int flat) => Clamp(flat) >= Top ? TopMajor : Clamp(flat) / SubPerMajor;
    public static int Sub(int flat) => Clamp(flat) >= Top ? 0 : Clamp(flat) % SubPerMajor;
    public static int Clamp(int flat) => flat < 0 ? 0 : flat > Top ? Top : flat;
    /// <summary>这一小阶集满几颗星就升；天魁星不封顶返回 0。对标原作，每一小阶都是五星。</summary>
    public static int StarsAt(int flat) => Major(flat) >= TopMajor ? 0 : MaxStars;
    public static string Name(int flat)
    {
        int m = Major(flat);
        return m >= TopMajor ? Majors[m] : Majors[m] + "·" + Subs[Sub(flat)];
    }
    public static string MajorName(int flat) => Majors[Major(flat)];
    /// <summary>局外排名分：升一小阶比任何星数都多。</summary>
    public static int Score(int flat, int stars) => Clamp(flat) * 100 + Math.Min(stars, 99) * 10;
    public static void Apply(ref int tier, ref int stars, bool win, out bool promoted, out bool demoted)
    {
        promoted = demoted = false;
        tier = Clamp(tier);
        if (stars < 0) stars = 0;
        int cap = StarsAt(tier);
        if (cap > 0 && stars > cap) stars = cap;
        if (win)
        {
            if (cap > 0 && stars + 1 >= cap)
            {
                tier++;
                stars = 0;
                promoted = true;
            }
            else stars++;
            return;
        }
        if (stars > 0) { stars--; return; }
        if (tier < Top && Sub(tier) > 0)
        {
            tier--;
            int back = StarsAt(tier);
            stars = back > 0 ? back - 1 : 0;
            demoted = true;
        }
    }
    /// <summary>旧版七阶存档（喽啰…天罡星）换算到新座次。</summary>
    public static int FromLegacy(int oldTier)
    {
        int[] map = { 0, 1, 4, 6, 7, 8, 10 };
        if (oldTier < 0) oldTier = 0;
        if (oldTier >= map.Length) oldTier = map.Length - 1;
        return map[oldTier] * SubPerMajor;
    }
}
```

```
}
// 文件: Assets/Scripts/Sim/Match.cs
using System;
using System.Collections.Generic;
namespace Rescue.Sim
{
    public sealed class Match
    {
        public readonly ChapterDef Chapter;
        public readonly Side[] Sides = new Side[2];
        public readonly List<SimEvent> Events = new List<SimEvent>();
        public float Time;
        public int Wave;
        public bool Over;
        /// <summary>新手引导时冻结时间，玩家仍可操作。</summary>
        public bool Hold;
        public int Winner = -1;
        readonly Rng _waveRng;
        readonly List<EnemyKind> _queue = new List<EnemyKind>();
        int _qi;
        float _spawnT;
        float _interval;
        float _crowdScale = 1f;
        float _nextWaveT = Tuning.FirstWaveAt;
        int _nextEnemyId = 1;
        public Side Player => Sides[0];
        public Side Rival => Sides[1];
        public Match(ChapterDef chapter, uint seed, Loadout playerItems = null)
        {
            Chapter = chapter;
            _waveRng = new Rng(seed * 2654435761u + 17);
            Sides[0] = new Side(0, this, false, seed * 7 + 1, playerItems);
            Sides[1] = new Side(1, this, true, seed * 13 + 5, WeaponLadder.ForRival(chapter, WeaponLadder.Power(playerItems)));
            Sides[1].IncomeMul = chapter.RivalIncome;
            if (chapter.MaxWaves > 0) _nextWaveT = 8.5f;
        }
        public bool FinalWaveSpawned => Chapter.MaxWaves > 0 && Wave >= Chapter.MaxWaves && _qi >= _queue.Count;
        public float NextWaveIn => _qi < _queue.Count || (Chapter.MaxWaves > 0 && Wave >= Chapter.MaxWaves) ? 0 : Math.Ma
            x(0, _nextWaveT - Time);
        public void Step(float dt)
        {
            if (Over || Hold) return;
            Time += dt;
            bool canStartNext = Chapter.MaxWaves <= 0 || Wave < Chapter.MaxWaves;
            if (canStartNext && _qi >= _queue.Count && Time >= _nextWaveT) StartWave();
            if (_qi < _queue.Count)
            {
                _spawnT -= dt;
                if (_spawnT <= 0)
                {
                    var kind = _queue[_qi++];
                    foreach (var s in Sides) SpawnOn(s, kind, 0);
```

```
    {
        var l = new Loadout { Rack = new List<int>(), RackLv = new List<int>() };
        foreach (var s in rack)
            foreach (var d in HeroDef.All)
            {
                if (d.WGrade != s.Grade || string.IsNullOrEmpty(d.Weapon) || d.NoAttack || d.Hidden) continue;
                if (Array.IndexOf(ch.Pool, d.Id) >= 0 || Weapons.MoveOf(d) != WeaponMove.None || l.Rack.Contains((int)d.Id)) conti
            nue;
                l.Rack.Add((int)d.Id);
                l.RackLv.Add(s.Tier);
                break;
            }
        return l;
    }
}
/// <summary>羁绊：同时凑出若干名有渊源的好汉，他们伤害提高。</summary>
public sealed class Bond
{
    public string Name, Story;
    public HeroId[] Members;
    public int Need;
    public float Dmg;
    public static readonly Bond[] All =
    {
        new Bond { Name = "阮氏三雄", Need = 2, Dmg = 0.25f, Story = "石碣村打鱼的三兄弟，同心智取生辰纲",
            Members = new[] { HeroId.RuanXiaoEr, HeroId.RuanXiaoWu, HeroId.RuanXiaoQi } },
        new Bond { Name = "五虎上将", Need = 2, Dmg = 0.15f, Story = "梁山马军五虎将",
            Members = new[] { HeroId.GuanSheng, HeroId.LinChong, HeroId.QinMing, HeroId.HuYanZhuo, HeroId.DongPing } },
        new Bond { Name = "八骠骑", Need = 2, Dmg = 0.12f, Story = "马军八骠骑兼先锋使",
            Members = new[] { HeroId.HuaRong, HeroId.XuNing, HeroId.YangZhi, HeroId.SuoChao, HeroId.ZhangQing, HeroId.Zh
            uTong, HeroId.ShiJin, HeroId.MuHong } },
        new Bond { Name = "二龙山", Need = 2, Dmg = 0.15f, Story = "鲁智深、杨志、武松同占二龙山",
            Members = new[] { HeroId.LuZhiShen, HeroId.YangZhi, HeroId.WuSong, HeroId.CaoZheng, HeroId.ShiEn, HeroId.Zhan
            gQingY, HeroId.SunErNiang } },
        new Bond { Name = "七星聚义", Need = 3, Dmg = 0.15f, Story = "晁盖与七人聚义，智取生辰纲",
            Members = new[] { HeroId.WuYong, HeroId.GongSunSheng, HeroId.LiuTang, HeroId.RuanXiaoEr, HeroId.RuanXiaoWu,
             HeroId.RuanXiaoQi, HeroId.BaiSheng } },
        new Bond { Name = "张氏兄弟", Need = 2, Dmg = 0.2f, Story = "浔阳江上船火儿与浪里白条",
            Members = new[] { HeroId.ZhangHeng, HeroId.ZhangShun } },
        new Bond { Name = "解氏兄弟", Need = 2, Dmg = 0.2f, Story = "登州打虎的两头蛇与双尾蝎",
            Members = new[] { HeroId.XieZhen, HeroId.XieBao } },
        new Bond { Name = "翠屏山", Need = 2, Dmg = 0.18f, Story = "杨雄、石秀结为兄弟，带时迁投梁山",
            Members = new[] { HeroId.YangXiong, HeroId.ShiXiu, HeroId.ShiQian } },
        new Bond { Name = "少华山", Need = 2, Dmg = 0.15f, Story = "史进与朱武、陈达、杨春不打不相识",
            Members = new[] { HeroId.ShiJin, HeroId.ZhuWu, HeroId.ChenDa, HeroId.YangChun } },
        new Bond { Name = "十字坡夫妻", Need = 2, Dmg = 0.18f, Story = "张青、孙二娘夫妻开店",
            Members = new[] { HeroId.ZhangQingY, HeroId.SunErNiang } },
        new Bond { Name = "王英扈三娘", Need = 2, Dmg = 0.18f, Story = "宋江作主，一丈青嫁了矮脚虎",
            Members = new[] { HeroId.WangYing, HeroId.HuSanNiang } },
        new Bond { Name = "登州孙家", Need = 2, Dmg = 0.15f, Story = "孙立、孙新、顾大嫂登州劫牢",
            Members = new[] { HeroId.SunLi, HeroId.SunXin, HeroId.GuDaSao, HeroId.YueHe } },
        new Bond { Name = "芒砀三杰", Need = 2, Dmg = 0.18f, Story = "混世魔王樊瑞与左右手项充、李袞",
```

```
            Members = new[] { HeroId.FanRui, HeroId.XiangChong, HeroId.LiGun } },
        new Bond { Name = "画戟双英", Need = 2, Dmg = 0.2f, Story = "吕方、郭盛对影山斗戟，花荣一箭射开",
            Members = new[] { HeroId.LvFang, HeroId.GuoSheng } },
        new Bond { Name = "揭阳水军", Need = 2, Dmg = 0.15f, Story = "李俊与童威、童猛浔阳江上做私商",
            Members = new[] { HeroId.LiJun, HeroId.TongWei, HeroId.TongMeng, HeroId.LiLi } },
        new Bond { Name = "铁牛与哥哥", Need = 2, Dmg = 0.2f, Story = "黑旋风生死追随宋江",
            Members = new[] { HeroId.SongJiang, HeroId.LiKui } },
        new Bond { Name = "主仆情深", Need = 2, Dmg = 0.2f, Story = "浪子燕青舍命救主卢俊义",
            Members = new[] { HeroId.LuJunYi, HeroId.YanQing } },
        new Bond { Name = "郓城都头", Need = 2, Dmg = 0.18f, Story = "朱仝、雷横先后私放晁盖、宋江",
            Members = new[] { HeroId.ZhuTong, HeroId.LeiHeng } },
        new Bond { Name = "水火二将", Need = 2, Dmg = 0.2f, Story = "凌州圣水将、神火将",
            Members = new[] { HeroId.ShanTingGui, HeroId.WeiDingGuo } },
        new Bond { Name = "江州好汉", Need = 2, Dmg = 0.12f, Story = "江州劫法场救宋江",
            Members = new[] { HeroId.DaiZong, HeroId.LiKui, HeroId.ZhangShun, HeroId.LiJun, HeroId.MuHong } },
    };
  }
}
// 文件: Assets/Scripts/Game/JoinPanel.cs
using System;
using System.Collections.Generic;
using Rescue.Sim;
using UnityEngine;
using UnityEngine.UI;
namespace Rescue
{
  /// <summary>赢下回目后的「好汉入伙」：一人一张拜帖，读完是本回的水浒小课堂；一百单八将聚齐时再放忠义堂排
    座次。</summary>
  public static class JoinPanel
  {
    static readonly Color Edge = Ui.Edge;
    static readonly Color TianColor = Pal.Hex("C8962E");
    static readonly Color DiColor = Pal.Hex("8A54C0");
    /// <summary>入伙流程里点了「查看好汉谱」：回到主页后直接打开好汉谱。</summary>
    public static bool WantCodex;
    public static void Show(List<HeroId> heroes, int chapter, bool seat, Action done)
    {
      Sfx.PlayStory();
      var layer = Ui.Fill(Ui.Root, new Color(0.07f, 0.05f, 0.04f, 0.92f), true, "join");
      int i = 0;
      RectTransform page = null;
      void Finish()
      {
        UnityEngine.Object.Destroy(layer.gameObject);
        if (seat) ShowSeats(done);
        else done?.Invoke();
      }
      void Lore()
      {
        if (page) UnityEngine.Object.Destroy(page.gameObject);
        var ch = chapter >= 0 && chapter < ChapterDef.StoryCount ? ChapterDef.All[chapter] : null;
        if (ch == null || string.IsNullOrEmpty(ch.Lore)) { Finish(); return; }
```

```
            page = LorePage(layer, ch, Finish);
        }
        void Next()
        {
            if (page) UnityEngine.Object.Destroy(page.gameObject);
            if (i >= heroes.Count) { Lore(); return; }
            var d = HeroDef.Get(heroes[i]);
            i++;
            page = Ui.Rect(Ui.Fit(layer, 1800), "page");
            Rays(page, new Vector2(0, 640));
            var title = Ui.Title(page, "好汉入伙", new Vector2(0, 820), 100, Pal.GoldBright);
            Tween.Pop(title.transform, 0.35f);
            Ui.Label(page, $"{i} / {heroes.Count}", 36, Pal.Paper.A(0.7f), new Vector2(0, 730), new Vector2(300, 50));
            var sheet = HeroSheet(page, d, true, new Vector2(0, -20));
            Tween.Pop(sheet, 0.4f, 0.05f);
            Sfx.Play(Snd.Join, 0.9f);
            bool last = i >= heroes.Count;
            Ui.CartoonBtn(page, last ? "继 续" : "下一位", new Vector2(420, 128), new Vector2(170, -760), Ui.BtnRed, Next, 56);
            Ui.CartoonBtn(page, "好汉谱", new Vector2(300, 112), new Vector2(-240, -760), Ui.BtnBlue, () =>
            {
                WantCodex = true;
                Next();
            }, 46);
        }
        Next();
    }
    static void Rays(RectTransform parent, Vector2 pos)
    {
        var clip = Ui.Rect(parent, "raysClip");
        clip.anchoredPosition = pos;
        clip.sizeDelta = new Vector2(1080, 700);
        var rays = Ui.Icon(clip, "ui_rays", 1000, Vector2.zero, "rays");
        rays.color = Color.white.A(0.55f);
        Ui.Spin(rays.transform, 10f);
    }
    /// <summary>
    /// 立绘窗。画布 640，脚在 0.86、头高的能到 0.25、兵器最宽约 0.58，把这段装进窗口并裁掉窗外。
    /// </summary>
    public static RectTransform Portrait(Transform parent, HeroId id, Vector2 pos, Vector2 window)
    {
        var clip = Ui.Rect(parent, "portrait");
        clip.anchoredPosition = pos;
        clip.sizeDelta = window;
        clip.gameObject.AddComponent<RectMask2D>();
        var ground = Ui.Box(clip, new Vector2(window.x * 0.52f, 16f), new Vector2(0, -window.y * 0.44f),
            new Color(0.22f, 0.13f, 0.06f, 0.32f), Art.Circle, "ground");
        ground.type = Image.Type.Simple;
        ground.raycastTarget = false;
        const float top = 0.23f, bot = 0.90f, wide = 0.62f;
        float canvas = Mathf.Min(window.y / (bot - top), window.x / wide);
        var fig = Ui.Box(clip, new Vector2(canvas, canvas), Vector2.zero, Color.white, null, "figure");
        fig.type = Image.Type.Simple;
```

```
            fig.preserveAspect = true;
            fig.raycastTarget = false;
            fig.enabled = false;
            fig.rectTransform.anchoredPosition = new Vector2(0f, ((top + bot) * 0.5f - 0.5f) * canvas);
            ArtPack.Ensure(Art.HeroPack(id), () =>
            {
                if (!fig) return;
                var sp = Art.HeroFrame(id, 0);
                if (sp == null) return;
                fig.sprite = sp;
                fig.enabled = true;
            });
            return clip;
        }
        /// <summary>好汉拜帖：星号座次、绰号姓名、职司、兵器（品级与特性）、事迹、本领、绝招、结局。好汉谱详情也用这张。</summary>
        public static RectTransform HeroSheet(Transform parent, HeroDef d, bool joined, Vector2 pos)
        {
            bool di = d.Disha;
            Color rim = d.Rank <= 0 ? Pal.Hex("8C6B4E") : di ? DiColor : TianColor;
            var holder = Ui.Rect(parent, "sheet");
            holder.anchoredPosition = pos;
            Ui.Box(holder, new Vector2(900, 1240), new Vector2(0, -14), Pal.Shadow.A(0.6f), null, "shadow");
            Ui.Framed(holder, new Vector2(880, 1220), Vector2.zero, Pal.Hex("F6EEDA"), rim, 10f, "card");
            Ui.Box(holder, new Vector2(840, 340), new Vector2(0, 390), rim.A(0.14f), null, "band").pixelsPerUnitMultiplier = 1f;
            var t = holder;
            if (joined) Portrait(t, d.Id, new Vector2(-230, 378), new Vector2(300, 300));
            if (d.Rank > 0)
            {
                Ui.Tag(t, d.Star, rim, new Vector2(-280, 560), 190, 34);
                Ui.Tag(t, (di ? "地煞 · " : "天罡 · ") + d.Rank108, rim, new Vector2(270, 560), 230, 30);
            }
            else Ui.Tag(t, "隐藏好汉", Pal.Hex("8C6B4E"), new Vector2(0, 560), 200, 32);
            if (joined)
            {
                Ui.Label(t, d.Nick, 40, Pal.Gold, new Vector2(188, 498), new Vector2(420, 56), TextAnchor.MiddleLeft, Fonts.Brush);
                var name = Ui.Label(t, d.Name, d.Name.Length > 2 ? 68 : 92, Pal.Ink, new Vector2(188, 400), new Vector2(420, 110), TextA
                nchor.MiddleLeft, Fonts.Brush);
                Ui.Stroke(name, Pal.Hex("FFF6E2"), 2.5f);
                if (!string.IsNullOrEmpty(d.Post))
                {
                    var post = Ui.Label(t, "梁山职司：" + d.Post, 30, Pal.InkSoft, new Vector2(188, 308), new Vector2(420, 72), TextAnchor.
                    MiddleLeft);
                    post.resizeTextForBestFit = true;
                    post.resizeTextMinSize = 20;
                    post.resizeTextMaxSize = 30;
                }
            }
            else
            {
                Ui.Label(t, d.Nick, 52, Pal.Gold, new Vector2(0, 470), new Vector2(800, 70), TextAnchor.MiddleCenter, Fonts.Brush);
                var name = Ui.Label(t, "？ ？ ", 190, Pal.Ink, new Vector2(0, 352), new Vector2(840, 210), TextAnchor.MiddleCenter, Fonts.
```

```
            Brush);
            Ui.Stroke(name, Pal.Hex("FFF6E2"), 3f);
            if (!string.IsNullOrEmpty(d.Post))
                Ui.Label(t, "梁山职司：" + d.Post, 34, Pal.InkSoft, new Vector2(0, 232), new Vector2(820, 46));
        }
        float y = 108;
        if (!string.IsNullOrEmpty(d.Weapon))
        {
            var g = Ui.GradeColor(d.WGrade);
            var plate = Ui.Framed(t, new Vector2(800, 150), new Vector2(0, y), Pal.Hex("EFE2C4"), g, 5f, "weapon");
            var pt = plate.transform;
            Ui.Box(pt, new Vector2(124, 124), new Vector2(-322, 0), g.A(0.25f), Art.Circle, "halo").type = Image.Type.Simple;
            Ui.Icon(pt, Weapons.Icon(d.WKind), 112, new Vector2(-322, 0));
            Ui.Stroke(Ui.Label(pt, d.Weapon, 50, Pal.Ink, new Vector2(-30, 30), new Vector2(420, 66), TextAnchor.MiddleLeft, Fonts.Br
            ush), Pal.Hex("FFF6E2"), 2f);
            Ui.Tag(pt, Weapons.Grades[Mathf.Clamp(d.WGrade, 0, 3)], g, new Vector2(300, 32), 120, 28);
            Ui.Label(pt, Weapons.KindName(d.WKind) + "  ·  " + Weapons.AffixText(d.WKind), 27, Pal.InkSoft, new Vector2(40, -34)
            , new Vector2(660, 44), TextAnchor.MiddleLeft);
            y -= 130;
        }
        else y -= 30;
        Ui.Label(t, d.Bio, 33, Pal.Ink, new Vector2(0, y - 90), new Vector2(800, 190), TextAnchor.UpperLeft);
        y -= 205;
        Ui.Label(t, $"<color=#B5651D><b>本领</b></color>    {d.Skill}", 31, Pal.Ink, new Vector2(0, y - 22), new Vector2(800, 5
            0), TextAnchor.MiddleLeft).supportRichText = true;
        y -= 60;
        if (!string.IsNullOrEmpty(d.Ult))
        {
            Ui.Label(t, $"<color=#B22222><b>绝招</b></color>    {d.Ult}", 31, Pal.Ink, new Vector2(0, y - 30), new Vector2(800, 8
            0), TextAnchor.UpperLeft).supportRichText = true;
            y -= 85;
        }
        Ui.Label(t, $"<color=#6B5A48><b>结局</b></color>    {d.Fate}", 29, Pal.InkSoft, new Vector2(0, y - 30), new Vector2(80
            0, 80), TextAnchor.UpperLeft).supportRichText = true;
        y -= 85;
        Ui.Label(t, "「" + d.Quote + "」", 34, Pal.RedDeep, new Vector2(0, Mathf.Min(y - 20, -500)), new Vector2(800, 56), TextAnch
            or.MiddleCenter, Fonts.Brush);
        int jc = ChapterDef.JoinChapter(d.Id);
        string debut = jc >= 0 ? $"第{BattleView.ToCn(jc + 1)}回「{ChapterDef.All[jc].Name}」入伙" : "隐藏：凑出名字即可得
            见";
        Ui.Label(t, debut, 28, Pal.InkSoft.A(0.85f), new Vector2(0, -570), new Vector2(800, 44));
        return holder;
    }
    static RectTransform LorePage(RectTransform layer, ChapterDef ch, Action done)
    {
        var page = Ui.Rect(Ui.Fit(layer, 1560), "lore");
        Ui.Title(page, "水浒小课堂", new Vector2(0, 700), 100, Pal.GoldBright);
        var card = Ui.WoodPanel(page, new Vector2(900, 900), new Vector2(0, 40), Pal.Hex("8C6B4E"), Pal.Hex("F3E6C6"));
        Ui.Label(card, $"第{BattleView.ToCn(ch.Index + 1)}回  ·  {ch.Name}", 52, Pal.Ink, new Vector2(0, 340), new Vector2(800, 80
            ), TextAnchor.MiddleCenter, Fonts.Brush);
        if (!string.IsNullOrEmpty(ch.Hui))
            Ui.Label(card, "《水浒传》" + ch.Hui, 32, Pal.Hex("9A6A1E"), new Vector2(0, 265), new Vector2(820, 50));
```

```
        Ui.Box(card, new Vector2(700, 4), new Vector2(0, 225), Pal.Hex("C9B48E"), null, "rule").pixelsPerUnitMultiplier = 1f;
        Ui.Label(card, ch.Lore, 42, Pal.Ink, new Vector2(0, 20), new Vector2(780, 380), TextAnchor.MiddleLeft);
        int next = ch.Index + 1;
        string tip = next < ChapterDef.StoryCount
            ? $"下一回：{ChapterDef.All[next].Name}（{ChapterDef.All[next].Act}）"
            : "剧情已终。聚义厅车轮战开张：自选五员大将，看能撑几阵！";
        Ui.Label(card, tip, 34, Pal.RedDeep, new Vector2(0, -260), new Vector2(800, 90));
        Ui.Label(card, $"好汉谱 {Save.JoinedCount} / 108", 32, Pal.InkSoft, new Vector2(0, -340), new Vector2(600, 50));
        Ui.CartoonBtn(page, "回山寨", new Vector2(420, 128), new Vector2(0, -560), Ui.BtnRed, done, 56);
        Tween.Pop(card, 0.35f);
        return page;
    }
    /// <summary>忠义堂排座次：石碣上刻满一百单八将，天罡金、地煞紫。</summary>
    public static void ShowSeats(Action done)
    {
        Sfx.PlayEpic();
        var layer = Ui.Fill(Ui.Root, new Color(0.05f, 0.04f, 0.03f, 0.95f), true, "seats");
        var fit = Ui.Fit(layer, 1880);
        Rays(fit, new Vector2(0, 760));
        var title = Ui.Title(fit, "英雄排座次", new Vector2(0, 860), 110, Pal.GoldBright);
        Tween.Pop(title.transform, 0.45f);
        Ui.Label(fit, "忠义堂前，天降石碣：替天行道 · 忠义双全", 38, Pal.Paper.A(0.85f), new Vector2(0, 765), new Vector2(1000
            , 56));
        Sfx.Play(Snd.RankUp);
        var stone = Ui.Framed(fit, new Vector2(1000, 1360), new Vector2(0, -40), Pal.Hex("4A4640"), Pal.Hex("8C8478"), 10f, "stele");
        var content = Ui.Scroll(stone.transform, new Vector2(980, 1340), Vector2.zero, 0);
        const int cols = 4;
        const float rowH = 64f;
        float h = 90 + 9 * rowH + 110 + 18 * rowH + 40;
        content.sizeDelta = new Vector2(980, h);
        float y = 50;
        Section(content, "三十六天罡", TianColor, ref y);
        int k = 0;
        foreach (var d in Ranked(1, 36)) { Seat(content, d, k++, cols, rowH, y, TianColor); }
        y += Mathf.Ceil(36f / cols) * rowH + 60;
        Section(content, "七十二地煞", DiColor, ref y);
        k = 0;
        foreach (var d in Ranked(37, 108)) { Seat(content, d, k++, cols, rowH, y, DiColor); }
        Ui.CartoonBtn(fit, "替天行道", new Vector2(460, 132), new Vector2(0, -800), Ui.BtnRed, () =>
        {
            UnityEngine.Object.Destroy(layer.gameObject);
            done?.Invoke();
        }, 58);
    }
    static IEnumerable<HeroDef> Ranked(int from, int to)
    {
        var list = new List<HeroDef>();
        foreach (var d in HeroDef.All) if (d.Rank >= from && d.Rank <= to) list.Add(d);
        list.Sort((a, b) => a.Rank.CompareTo(b.Rank));
        return list;
    }
    static void Section(RectTransform content, string name, Color c, ref float y)
```

```
        {
            var l = Ui.Label(content, name, 52, c, Ui.Top(content, 0, y), new Vector2(900, 70), TextAnchor.MiddleCenter, Fonts.Brush);
            Ui.Stroke(l, Edge, 2.5f);
            y += 60;
        }
        static void Seat(RectTransform content, HeroDef d, int k, int cols, float rowH, float y0, Color c)
        {
            float w = 940f / cols;
            float x = -470 + w * (k % cols + 0.5f);
            float y = y0 + (k / cols) * rowH + rowH * 0.5f;
            var l = Ui.Label(content, $"<size=22><color=#BFB7A8>{d.Rank}</color></size> <color=#{ColorUtility.ToHtmlString
                RGB(c)}>{d.Nick}</color> {d.Name}",
                28, Pal.Paper, Ui.Top(content, x, y), new Vector2(w, rowH), TextAnchor.MiddleCenter);
            l.supportRichText = true;
            l.horizontalOverflow = HorizontalWrapMode.Overflow;
        }
    }
}
// 文件: Assets/Scripts/Game/ResultPanel.cs
using System.Collections.Generic;
using Rescue.Sim;
using UnityEngine;
using UnityEngine.UI;
namespace Rescue
{
    /// <summary>结算：胜利是红旗+光芒+卷轴星+众人起舞+双倍金币；失败是乌云+掉星+被围的宋江，可看广告保星。
        缴获的兵器不铺在这一页上，领完再单独呈上。</summary>
    public static class ResultPanel
    {
        static readonly Color Edge = Ui.Edge;
        public static int WinCoins(ChapterDef ch) => WinCoins(ch.Index, ch.Round);
        public static int WinCoins(int chapter, int round) => Economy.WinCoins(chapter, round);
        /// <summary>每日一战：当天首胜给大头，之后再赢只给一点。</summary>
        public static int DailyWinCoins => Save.DailyWonToday ? 15 : Save.DailyCoinsReward;
        /// <summary>preview 有值时，结算一定演出这件兵器落下。截图用，正式对局不传。</summary>
        public static void Show(BattleResult r, System.Action onClose, HeroId? preview = null)
        {
            bool dailyFirst = false;
            int coins;
            if (r.Daily)
            {
                coins = r.Quit ? 0 : r.Win ? (Save.DailyWonToday ? 15 : 0) : Economy.LoseCoins(r.Wave);
                if (r.Win) dailyFirst = Save.WinDaily();
                if (dailyFirst) r.SureDrop = true;
                if (!r.Quit) Analytics.DailyBattle(r.Win, dailyFirst);
            }
            else coins = r.Quit ? 0 : (r.Win ? WinCoins(r.Chapter, r.Round) : Economy.LoseCoins(r.Wave));
            var rank = r.Quit || r.Daily
                ? new Save.RankDelta { OldTier = Save.D.tier, OldStars = Save.D.stars }
                : Save.ApplyResult(r.Win);
            if (!r.Quit && !r.Daily) Analytics.RankChange(rank.OldTier, rank.OldStars, Save.D.tier, Save.D.stars, r.Win);
            var joined = new List<HeroId>();
```

```
        if (r.Daily) joined.Clear();
        else if (r.Round > 0)
        {
            if (r.Win)
            {
                Save.D.endlessBest = Mathf.Max(Save.D.endlessBest, r.Round);
                Save.D.endlessRound = r.Round + 1;
            }
            else if (!r.Quit) Save.D.endlessRound = 1;
        }
        else if (r.Win) joined = Save.WinStory(r.Chapter);
        bool seatNow = joined.Count > 0 && !Save.D.seated && Save.AllSeated;
        if (seatNow) Save.D.seated = true;
        foreach (var id in r.Formed) if (!Save.D.heroes.Contains((int)id)) Save.D.heroes.Add((int)id);
        Save.D.coins += coins;
        if (dailyFirst) coins += Save.DailyCoinsReward;
        var loot = Save.RollWeapon(r);
        if (!string.IsNullOrEmpty(loot.Text)) Analytics.WeaponDrop(HeroDef.Get(loot.Hero).Weapon, HeroDef.Get(loot.Hero).WGr
            ade, !loot.IsWeapon);
        bool showWeapon = loot.IsWeapon;
        HeroId dropped = loot.Hero;
        if (preview is HeroId forced)
        {
            showWeapon = true;
            dropped = forced;
        }
        Save.Write();
        LiveConfig.SubmitRank();
        var layer = Ui.Fill(Ui.Root, r.Win ? Pal.Hex("F3EAD6") : Pal.Hex("C9D6DC"), true, "result");
        var c = Ui.Rect(Ui.Fit(layer, 1880), "c");
        bool claimed = false;
        void Close()
        {
            if (claimed) return;
            claimed = true;
            Object.Destroy(layer.gameObject);
            void Next()
            {
                if (joined.Count > 0) JoinPanel.Show(joined, r.Chapter, seatNow, onClose);
                else onClose?.Invoke();
            }
            if (showWeapon) ShowLoot(HeroDef.Get(dropped), Next);
            else Next();
        }
        if (r.Win) Victory(c, rank);
        else Defeat(c, rank, r.Quit);
        var banner = BuildScroll(c, rank, r.Win, out var stars);
        var stage = BuildStage(c, r.Win);
        var tags = new List<(string, Color)>();
        if (dailyFirst) tags.Add(("每日一战 首胜", Pal.Hex("C0392B")));
        else if (r.Daily) tags.Add(("每日一战 不动座次", Pal.Hex("6E5D45")));
        if (showWeapon) tags.Add(("缴获 " + HeroDef.Get(dropped).Weapon, Ui.GradeColor(HeroDef.Get(dropped).WGrade)));
```

```
        else if (!string.IsNullOrEmpty(loot.Text)) tags.Add((loot.Text, Pal.Hex("8A54C0")));
        if (joined.Count > 0) tags.Add(($"{joined.Count} 位好汉入伙", Pal.Hex("C0392B")));
        foreach (var id in r.NewHeroes) tags.Add(("初次聚义 " + HeroDef.Get(id).Name, Pal.Hex("C8962E")));
        float tx = -(tags.Count - 1) * 170f;
        for (int i = 0; i < tags.Count && i < 3; i++)
        {
            var tag = Ui.Tag(c, tags[i].Item1, tags[i].Item2, new Vector2(tx + i * 340, -262), 0, 30);
            Tween.Pop(tag.transform.parent, 0.3f, 1.0f + i * 0.12f);
        }
        Ui.Label(c, r.Quit ? "撤回山寨，不记座次" : $"撑到第 {r.Wave} 波　斩敌 {r.Kills}　及时雨 {r.Rains} 次", 32, Pal.InkSoft,
            new Vector2(0, -195), new Vector2(900, 50));
        Text coinText = null;
        if (coins > 0)
        {
            var row = Ui.Rect(c, "coins");
            row.anchoredPosition = new Vector2(0, -350);
            Ui.Icon(row, "coin", 96, new Vector2(-70, 0));
            coinText = Ui.Label(row, "×" + coins, 64, Pal.Hex("F2B53A"), new Vector2(70, 0), new Vector2(220, 90), TextAnchor.Midd
            leLeft);
            coinText.fontStyle = FontStyle.Bold;
            Ui.Stroke(coinText, Edge, 3f);
            Tween.Pop(row, 0.35f, 1.1f);
            CoinBurst(c, new Vector2(0, -350), 1.15f);
        }
        float y = -500;
        if (coins > 0)
        {
            RectTransform dbl = null;
            dbl = Ui.CartoonBtn(c, "双倍金币", new Vector2(560, 132), new Vector2(0, y), Ui.BtnRed, () =>
                AdStub.Show("双倍铜钱", () =>
                {
                    Save.D.coins += coins;
                    Save.Write();
                    coinText.text = "×" + coins * 2;
                    Tween.Pop(coinText.transform, 0.3f);
                    CoinBurst(c, new Vector2(0, -350), 0f);
                    Sfx.Play(Snd.Rain);
                    if (dbl) dbl.gameObject.SetActive(false);
                    Tween.Run(coinText, 1.0f, k => { }, Close);
                }), 58, "ui_video");
            AdStub.Badge(dbl, new Vector2(250, 60), "双倍铜钱");
            Breathe(dbl);
            y -= 160;
        }
        if (!r.Win && !r.Quit && (rank.DroppedStar || rank.Demoted))
        {
            RectTransform keep = null;
            keep = Ui.CartoonBtn(c, "保住星星", new Vector2(560, 132), new Vector2(0, y), Ui.BtnOrange, () =>
                AdStub.Show("保星", () =>
                {
                    Save.D.tier = rank.OldTier;
                    Save.D.stars = rank.OldStars;
```

```
                Save.Write();
                stars.text = Ui.RankStars(Save.D.tier, Save.D.stars);
                Tween.Pop(stars.transform, 0.35f);
                Sfx.Play(Snd.Merge);
                Ui.Toast("星星保住了！");
                if (keep) keep.gameObject.SetActive(false);
              }), 58, "ui_video");
            AdStub.Badge(keep, new Vector2(250, 60), "保星");
            Breathe(keep);
            y -= 160;
        }
        Ui.CartoonBtn(c, r.Win ? "领取" : (coins > 0 ? "领取" : "关闭"), new Vector2(380, 112), new Vector2(0, y), Ui.BtnDark, Close,
            52);
        string next = showWeapon && joined.Count > 0 ? "领完铜钱，再看缴获的兵器，好汉来拜山"
            : showWeapon ? "领完铜钱，再看缴获的兵器"
            : joined.Count > 0 ? "领完铜钱，新入伙的好汉来拜山"
            : "领完铜钱，孙二娘的摊子就摆开";
        Ui.Label(c, next, 28, Pal.InkSoft.A(0.8f), new Vector2(0, y - 100), new Vector2(800, 44));
        Tween.Pop(banner, 0.4f, 0.15f);
        Tween.Pop(stage, 0.4f, 0.3f);
    }
    /// <summary>缴获单独一页。兵器本身是主体，背后只留胜利同款的光，不再套方框圆框。</summary>
    public static void ShowLoot(HeroDef d, System.Action done)
    {
        int grade = Mathf.Clamp(d.WGrade, 0, 3);
        var g = Ui.GradeColor(grade);
        bool relic = grade >= 3;
        var layer = Ui.Fill(Ui.Root, Pal.Hex("F3EAD6"), true, "loot");
        var page = Ui.Rect(Ui.Fit(layer, 1860), "page");
        const float land = 130f;
        float iconSize = relic ? 860f : 780f;
        // 没有描边的光。ui_rays 外圈是黑线，整圆露出来就像劣质徽章。
        var wash = Ui.Box(page, Vector2.one * (relic ? 1680f : 1500f), new Vector2(0, land + 30f), Pal.Hex("F6C531").A(relic ? 0.5f : 0.
            34f), Art.SoftDot, "wash");
        wash.type = Image.Type.Simple;
        var core = Ui.Box(page, Vector2.one * (relic ? 980f : 860f), new Vector2(0, land + 10f), Pal.Hex("FFF8E4").A(relic ? 0.85f : 0.62
            f), Art.SoftDot, "core");
        core.type = Image.Type.Simple;
        var flag = Ui.Icon(page, "ui_flag", 440, new Vector2(6, 820), "flag");
        var ft = Ui.Label(flag.transform, "缴获", 78, Pal.Paper, new Vector2(6, 12), new Vector2(320, 110), TextAnchor.MiddleCenter,
            Fonts.Brush);
        Ui.Stroke(ft, Pal.Hex("7A1F12"), 4f);
        Wave(flag.transform);
        Tween.Pop(flag.transform, 0.4f);
        var shadow = Ui.Box(page, new Vector2(520, 520), new Vector2(0, land - 340f), new Color(0.25f, 0.16f, 0.08f, 0f), Art.SoftDot,
            "shadow");
        shadow.type = Image.Type.Simple;
        shadow.rectTransform.localScale = new Vector3(0.3f, 0.12f, 1f);
        var icon = Ui.Icon(page, Weapons.Icon(d.WKind), iconSize, new Vector2(0, land + 240f), "weapon");
        int nameSize = d.Weapon.Length > 5 ? 76 : 112;
        var name = Ui.Label(page, d.Weapon, nameSize, Pal.Ink, new Vector2(0, -400), new Vector2(980, 150), TextAnchor.MiddleCe
            nter, Fonts.Brush);
```

```
        Ui.Stroke(name, relic ? Pal.Hex("F2C14E") : Pal.Hex("FFF6E2"), relic ? 4f : 3f);
        name.transform.localScale = Vector3.zero;
        string who = string.IsNullOrEmpty(d.Nick) ? d.Name : d.Nick + "  ·  " + d.Name;
        string hex = ColorUtility.ToHtmlStringRGB(g);
        var owner = Ui.Label(page, who + "    <color=#" + hex + ">" + Weapons.Grades[grade] + "</color>", 40, Pal.InkSoft, new
            Vector2(0, -510), new Vector2(900, 56), TextAnchor.MiddleCenter, Fonts.Brush);
        owner.supportRichText = true;
        owner.transform.localScale = Vector3.zero;
        var move = Weapons.MoveOf(d);
        string affixText = Weapons.AffixText(d.WKind);
        if (move != WeaponMove.None) affixText += $"\n<color=#B0451F>名兵招式 · {Weapons.MoveName(move)}</color>:
            {Weapons.MoveText(move)}";
        var affix = Ui.Label(page, affixText, move != WeaponMove.None ? 32 : 36, Pal.Ink, new Vector2(0, -590), new Vector2(940, 10
            0));
        affix.supportRichText = true;
        affix.transform.localScale = Vector3.zero;
        var give = Ui.Label(page, "挂上兵器架，凑出好汉就自动领", 32, Pal.InkSoft, new Vector2(0, -670), new Vector2(860, 48));
        give.transform.localScale = Vector3.zero;
        Ui.CartoonBtn(page, "收入囊中", new Vector2(460, 128), new Vector2(0, -820), relic ? Ui.BtnOrange : Ui.BtnRed, () =>
        {
            Object.Destroy(layer.gameObject);
            done?.Invoke();
        }, 52);
        Tween.Run(icon, 0.48f, k =>
        {
            if (!icon) return;
            float e = k * k;
            float y = Mathf.Lerp(land + 240f, land, e);
            float sway = Mathf.Sin(k * Mathf.PI * 2f) * 16f * (1f - k);
            icon.rectTransform.anchoredPosition = new Vector2(sway, y);
            icon.rectTransform.localRotation = Quaternion.Euler(0, 0, sway * 0.45f);
            icon.rectTransform.localScale = Vector3.one * Mathf.Lerp(0.72f, 1f, e);
            if (shadow)
            {
                shadow.rectTransform.localScale = new Vector3(Mathf.Lerp(0.35f, 1.15f, e), 0.12f, 1f);
                shadow.color = new Color(0.25f, 0.16f, 0.08f, Mathf.Lerp(0f, 0.45f, e));
            }
        }, () =>
        {
            if (!icon) return;
            icon.rectTransform.anchoredPosition = new Vector2(0, land);
            icon.rectTransform.localRotation = Quaternion.identity;
            Tween.Run(icon, 0.22f, k =>
            {
                if (!icon) return;
                float hop = Mathf.Sin(k * Mathf.PI);
                icon.rectTransform.anchoredPosition = new Vector2(0, land + hop * 22f);
                icon.rectTransform.localScale = new Vector3(1f + hop * 0.06f, 1f - hop * 0.1f, 1);
            }, () =>
            {
                if (!icon) return;
                icon.rectTransform.localScale = Vector3.one;
```

```
            icon.rectTransform.anchoredPosition = new Vector2(0, land);
        });
        Tween.Pop(name.transform, 0.32f, 0.02f);
        Tween.Pop(owner.transform, 0.28f, 0.08f);
        Tween.Pop(affix.transform, 0.28f, 0.14f);
        Tween.Pop(give.transform, 0.28f, 0.18f);
        Sfx.Play(relic ? Snd.RankUp : Snd.Hero, 0.9f);
        int n = relic ? 12 : 8;
        for (int i = 0; i < n; i++)
        {
            float ang = i / (float)n * Mathf.PI * 2f;
            var sp = Ui.Icon(page, "ui_sparkle", relic ? 80 : 60, new Vector2(0, land), "sp");
            var dir = new Vector2(Mathf.Cos(ang), Mathf.Sin(ang)) * (relic ? 260f : 200f);
            Tween.Run(sp, 0.45f, k =>
            {
                if (!sp) return;
                sp.rectTransform.anchoredPosition = new Vector2(0, land) + dir * Tween.OutCubic(k);
                sp.color = Color.white.A(1f - k);
            }, () => { if (sp) Object.Destroy(sp.gameObject); });
        }
    }, 0.2f);
}
static void Breathe(RectTransform t)
{
    Tween.Run(t, 1.2f, k => { if (t) t.localScale = Vector3.one * (1f + Mathf.Sin(k * Mathf.PI * 2) * 0.04f); }, () => { if (t) Breathe(
        t); });
}
static void Victory(RectTransform c, Save.RankDelta rank)
{
    var clip = Ui.Rect(c, "raysClip");
    clip.anchoredPosition = new Vector2(0, 620);
    clip.sizeDelta = new Vector2(1080, 480);
    clip.gameObject.AddComponent<RectMask2D>();
    var rays = Ui.Icon(clip, "ui_rays", 860, new Vector2(0, -240), "rays");
    Ui.Spin(rays.transform, 14f);
    Tween.Pop(rays.transform, 0.5f);
    var flag = Ui.Icon(c, "ui_flag", 560, new Vector2(10, 760), "flag");
    var ft = Ui.Label(flag.transform, "胜利", 92, Pal.Paper, new Vector2(10, 20), new Vector2(400, 130), TextAnchor.MiddleCente
        r, Fonts.Brush);
    Ui.Stroke(ft, Pal.Hex("7A1F12"), 4f);
    Wave(flag.transform);
    Tween.Pop(flag.transform, 0.45f);
    string head = rank.Promoted ? "恭喜晋升" : Save.D.stars > rank.OldStars ? "升 星" : "大 胜";
    var h = Ui.Label(c, head, 70, Pal.Hex("FFE36A"), new Vector2(0, 602), new Vector2(700, 100), TextAnchor.MiddleCenter, Fon
        ts.Brush);
    Ui.Stroke(h, Edge, 4f);
    Tween.Pop(h.transform, 0.4f, 0.3f);
}
static void Defeat(RectTransform c, Save.RankDelta rank, bool quit)
{
    var cloud = Ui.Icon(c, "ui_cloud", 560, new Vector2(0, 700), "cloud");
    var ct = Ui.Label(cloud.transform, quit ? "撤退" : "失败", 96, Pal.Paper, new Vector2(0, 30), new Vector2(400, 130), TextAnch
```

```
            or.MiddleCenter, Fonts.Brush);
        Ui.Stroke(ct, Edge, 4f);
        Tween.Pop(cloud.transform, 0.45f);
        var basePos = cloud.rectTransform.anchoredPosition;
        Tween.Run(cloud, 3f, k => { if (cloud) cloud.rectTransform.anchoredPosition = basePos + new Vector2(Mathf.Sin(k * Mathf.
            PI * 2) * 14f, 0); });
    }
    /// <summary>挂在中间的黑布卷轴：座次名 + 五颗星。掉的那颗星先亮着再碎成灰，旁边红箭头往下。</summary>
    static RectTransform BuildScroll(RectTransform c, Save.RankDelta rank, bool win, out Text stars)
    {
        var holder = Ui.Rect(c, "scroll");
        holder.anchoredPosition = new Vector2(0, 410);
        var sc = Ui.Icon(holder, "ui_scroll", 820, Vector2.zero);
        sc.rectTransform.sizeDelta = new Vector2(820, 300);
        var plate = Ui.Framed(holder, new Vector2(380, 70), new Vector2(0, 118), Pal.Hex("2B2622"), Ui.RankColor(Save.D.tier), 4f, "
            plate");
        Ui.Stroke(Ui.Label(plate.transform, Save.RankName, 44, Pal.Hex("FFE36A"), new Vector2(0, 2), new Vector2(380, 70), TextA
            nchor.MiddleCenter, Fonts.Brush), Edge, 2f);
        bool tierChanged = rank.Promoted || rank.Demoted;
        int cap = Mathf.Max(1, RankRules.StarsAt(Save.D.tier));
        int from = tierChanged ? (rank.Promoted ? 0 : cap) : rank.OldStars;
        int to = Save.D.stars;
        int tierNow = Save.D.tier;
        stars = Ui.Label(holder, Ui.RankStars(tierNow, from), cap >= 5 ? 104 : 116, Color.white, new Vector2(0, -20), new Vector2(72
            0, 150));
        stars.supportRichText = true;
        Ui.Stroke(stars, Edge, 3f);
        var st = stars;
        if (from != to)
        {
            Tween.Run(st, 0.01f, k => { }, () =>
            {
                if (!st) return;
                st.text = Ui.RankStars(tierNow, to);
                Tween.Punch(st.transform, 0.35f, 0.18f);
                Sfx.Play(win ? Snd.Merge : Snd.Lose, 0.6f);
            }, 0.9f);
            int changed = win ? Mathf.Clamp(to - 1, 0, cap - 1) : Mathf.Clamp(from - 1, 0, cap - 1);
            float step = cap >= 5 ? 104 : 116;
            float sx = (changed - (cap - 1) * 0.5f) * step;
            if (win)
            {
                var spark = Ui.Icon(holder, "ui_sparkle", 150, new Vector2(sx, -20), "spark");
                spark.transform.localScale = Vector3.zero;
                Tween.Run(spark, 0.6f, k => { if (spark) { spark.transform.localScale = Vector3.one * Mathf.Sin(k * Mathf.PI) * 1.2f; spa
            rk.transform.localRotation = Quaternion.Euler(0, 0, k * 120f); } }, null, 0.9f);
            }
            else
            {
                var arrow = Ui.Label(holder, " ↓ ", 90, Pal.Hex("E8392B"), new Vector2(sx + 50, 50), new Vector2(80, 110));
                Ui.Stroke(arrow, Pal.Paper, 2f);
                arrow.color = Color.clear;
```

```
            var a0 = arrow.rectTransform.anchoredPosition;
            Tween.Run(arrow, 0.8f, k =>
            {
                if (!arrow) return;
                arrow.color = Pal.Hex("E8392B");
                arrow.rectTransform.anchoredPosition = a0 + new Vector2(0, -Mathf.Repeat(k * 2f, 1f) * 26f);
            }, null, 0.9f);
        }
    }
    return holder;
}
/// <summary>中间的草席小台：胜了宋江戴着幞头，周围一圈「舞」字蹦跳；输了宋江歪倒，被「兵」字围着。</summary>
static RectTransform BuildStage(RectTransform c, bool win)
{
    var holder = Ui.Rect(c, "stage");
    holder.anchoredPosition = new Vector2(0, 60);
    var mat = Ui.Icon(holder, "home_board", 600, new Vector2(0, -60));
    mat.rectTransform.sizeDelta = new Vector2(600, 320);
    if (!win) mat.color = Pal.Hex("B9C3C8");
    var song = Ui.Label(holder, "宋", 170, Pal.Ink, new Vector2(0, -10), new Vector2(220, 220), TextAnchor.MiddleCenter, Fonts.Brush);
    Ui.Stroke(song, Pal.Paper, 2.5f);
    var hat = Ui.Icon(holder, "hat", 110, new Vector2(4, 120));
    if (win)
    {
        Hop(song.transform, 0f, 16f);
        Hop(hat.transform, 0.05f, 16f);
        var spots = new[] { new Vector2(-300, 90), new Vector2(300, 90), new Vector2(-210, -120), new Vector2(220, -120) };
        for (int i = 0; i < spots.Length; i++)
        {
            var w = Ui.Label(holder, "舞", 110, Pal.Ink, spots[i], new Vector2(150, 150), TextAnchor.MiddleCenter, Fonts.Brush);
            Ui.Stroke(w, Pal.Paper, 2f);
            Dance(w.transform, i * 0.25f, i % 2 == 0 ? 1 : -1);
            var ribbon = Ui.Box(holder, new Vector2(110, 10), spots[i] + new Vector2(i % 2 == 0 ? 70 : -70, 30), Pal.Hex("E05A8A"), Art.Stroke, "ribbon");
            ribbon.type = Image.Type.Simple;
            ribbon.transform.SetSiblingIndex(1);
            Dance(ribbon.transform, i * 0.25f + 0.1f, i % 2 == 0 ? -1 : 1);
        }
    }
    else
    {
        song.color = Pal.InkSoft;
        song.rectTransform.localRotation = Quaternion.Euler(0, 0, -18);
        hat.rectTransform.anchoredPosition = new Vector2(120, -110);
        hat.rectTransform.localRotation = Quaternion.Euler(0, 0, 70);
        var spots = new[] { new Vector2(-280, 80), new Vector2(280, 80), new Vector2(-230, -130), new Vector2(240, -140), new Vector2(-60, 160) };
        for (int i = 0; i < spots.Length; i++)
        {
            var b = Ui.Label(holder, "兵", 96, Pal.Ink, spots[i], new Vector2(130, 130), TextAnchor.MiddleCenter, Fonts.Brush);
```

```
            Ui.Stroke(b, Pal.Paper, 2f);
            Jitter(b.transform, i * 0.2f);
        }
        var sweat = Ui.Label(holder, "；", 60, Pal.Hex("5A8FC0"), new Vector2(100, 80), new Vector2(80, 80));
        Hop(sweat.transform, 0.3f, 8f);
    }
    return holder;
}
static void Wave(Transform t)
{
    Tween.Run(t, 1.6f, k => { if (t) t.localRotation = Quaternion.Euler(0, 0, Mathf.Sin(k * Mathf.PI * 2) * 3f); }, () => { if (t) Wav
        e(t); });
}
static void Hop(Transform t, float phase, float h)
{
    var rt = (RectTransform)t;
    var p0 = rt.anchoredPosition;
    Tween.Run(t, 0.7f, k => { if (t) rt.anchoredPosition = p0 + new Vector2(0, Mathf.Abs(Mathf.Sin((k + phase) * Mathf.PI)) * h);
        },
        () => { if (t) { rt.anchoredPosition = p0; Hop(t, phase, h); } });
}
static void Dance(Transform t, float phase, int dir)
{
    var rt = (RectTransform)t;
    var p0 = rt.anchoredPosition;
    Tween.Run(t, 0.9f, k =>
    {
        if (!t) return;
        float s = Mathf.Sin((k + phase) * Mathf.PI * 2);
        rt.anchoredPosition = p0 + new Vector2(s * 10f * dir, Mathf.Abs(s) * 22f);
        rt.localRotation = Quaternion.Euler(0, 0, s * 16f * dir);
    }, () => { if (t) { rt.anchoredPosition = p0; Dance(t, phase, dir); } });
}
static void Jitter(Transform t, float phase)
{
    var rt = (RectTransform)t;
    var p0 = rt.anchoredPosition;
    Tween.Run(t, 0.5f, k =>
    {
        if (!t) return;
        float s = Mathf.Sin((k + phase) * Mathf.PI * 2);
        rt.anchoredPosition = p0 + new Vector2(s * 6f, 0);
        rt.localRotation = Quaternion.Euler(0, 0, s * 6f);
    }, () => { if (t) { rt.anchoredPosition = p0; Jitter(t, phase); } });
}
/// <summary>铜钱从中间往外蹦一圈再落回数字。</summary>
static void CoinBurst(RectTransform c, Vector2 at, float delay)
{
    for (int i = 0; i < 12; i++)
    {
        var coin = Ui.Icon(c, "coin", 64, at, "burst");
        coin.transform.localScale = Vector3.zero;
```

```
            float ang = i / 12f * Mathf.PI * 2f + 0.3f;
            var dir = new Vector2(Mathf.Cos(ang), Mathf.Sin(ang) * 0.8f + 0.4f) * (180 + (i % 3) * 50);
            var cc = coin;
            Tween.Run(cc, 0.9f, k =>
            {
              if (!cc) return;
              float up = k < 0.45f ? Tween.OutCubic(k / 0.45f) : 1f - (k - 0.45f) / 0.55f;
              cc.rectTransform.anchoredPosition = at + dir * up;
              cc.transform.localScale = Vector3.one * (k < 0.85f ? 1f : (1f - k) / 0.15f);
              cc.transform.localRotation = Quaternion.Euler(0, 0, k * 540f);
            }, () => { if (cc) Object.Destroy(cc.gameObject); }, delay + i * 0.02f);
          }
        }
      }
    }
    // 文件: cloudfunctions/songjiang-api/index.js
    /**
     * 营救宋江后端（CloudBase 事件云函数 + HTTP 访问），照 huahua-api。
     * gameKey 与经分一致：songjiang。
     *
     *   POST /login
     *   POST /save/pull
     *   POST /save/push
     *   POST /config       服务器时间、公告、广告位和各入口次数、按 IP 定的省份
     *   POST /rank/submit  上报座次分（省份由服务器按 IP 定）
     *   POST /rank/list    省榜 / 全国榜
     *   POST /avatar       把微信头像转回来，真机不用直连头像域名
     *   GET|POST /health
     *
     * 环境变量：GAME_KEY=songjiang，SONGJIANG_JWT_SECRET 必填。
     * 微信真机再配 SONGJIANG_WX_APPID / SONGJIANG_WX_SECRET。
     * 集合：songjiang_playerData（userId 唯一索引）、songjiang_rank（userId 唯一）、songjiang_config（_id=live）
     */
    const { handleLogin } = require('./lib/auth');
    const { handlePull, handlePush } = require('./lib/save');
    const { handleConfig, handleRankSubmit, handleRankList, handleAvatar } = require('./lib/live');
    const { respond, parseEvent, preflight } = require('./lib/http');
    const ROUTES = {
      'GET /health': async () => ({ ok: true, ts: Date.now() }),
      'POST /health': async () => ({ ok: true, ts: Date.now() }),
      'POST /login': handleLogin,
      'POST /save/pull': handlePull,
      'POST /save/push': handlePush,
      'POST /config': handleConfig,
      'GET /config': handleConfig,
      'POST /rank/submit': handleRankSubmit,
      'POST /rank/list': handleRankList,
      'POST /avatar': handleAvatar,
    };
    exports.main = async (event, context) => {
      try {
        if (event && event.httpMethod === 'OPTIONS') return preflight();
```

```
    const req = parseEvent(event);
    const key = `${req.method} ${req.path}`;
    const handler = ROUTES[key];
    if (!handler) {
      return respond(404, { ok: false, code: 'NOT_FOUND', error: `no route: ${key}` });
    }
    const result = await handler(req, context);
    if (result && typeof result === 'object' && 'statusCode' in result) return result;
    return respond(200, { ok: true, data: result });
  } catch (error) {
    const code = error && error.code ? error.code : 'INTERNAL';
    const status = error && error.status ? error.status : 500;
    const message = (error && error.message) || String(error);
    console.error('[songjiang-api] error:', code, message, error && error.stack);
    const out = { ok: false, code, error: message };
    if (error && error.data !== undefined) out.data = error.data;
    return respond(status, out);
  }
};
// 文件: cloudfunctions/songjiang-api/lib/config.js
const DEFAULT_GAME_KEY = 'songjiang';
const DEFAULT_TTL_SEC = 7 * 24 * 3600;
const DEFAULT_MAX_BYTES = 256 * 1024;
function getGameKey() {
  const v = String(process.env.GAME_KEY || '').trim().toLowerCase();
  if (!v) return DEFAULT_GAME_KEY;
  if (!/^[a-z][a-z0-9_\-]{0,31}$/.test(v)) {
    throw new Error(`非法 GAME_KEY: ${v}`);
  }
  return v;
}
function gameKeyUpper() {
  return getGameKey().toUpperCase().replace(/[^A-Z0-9]/g, '_');
}
function readEnvPrefer(...keys) {
  for (const key of keys) {
    const value = process.env[key];
    if (value !== undefined && value !== null && String(value).length > 0) {
      return String(value);
    }
  }
  return '';
}
function getCollectionName(suffix) {
  const overrideKey = `${gameKeyUpper()}_${suffix.toUpperCase()}_COLLECTION`;
  const override = process.env[overrideKey];
  if (override) return String(override);
  return `${getGameKey()}_${suffix}`;
}
function getJwtSecret() {
  return readEnvPrefer(`${gameKeyUpper()}_JWT_SECRET`);
}
```

```
function getTtlSec() {
  const raw = readEnvPrefer(`${gameKeyUpper()}_TOKEN_TTL_SEC`);
  const v = Number(raw);
  return Number.isFinite(v) && v > 0 ? Math.floor(v) : DEFAULT_TTL_SEC;
}
function getMaxBytes() {
  const raw = readEnvPrefer(`${gameKeyUpper()}_SAVE_MAX_BYTES`);
  const v = Number(raw);
  return Number.isFinite(v) && v > 0 ? Math.floor(v) : DEFAULT_MAX_BYTES;
}
function getPlatformCredential(platform, field) {
  const upper = gameKeyUpper();
  const platformUpper = platform.toUpperCase();
  return readEnvPrefer(`${upper}_${platformUpper}_${field}`);
}
module.exports = {
  getGameKey,
  gameKeyUpper,
  readEnvPrefer,
  getCollectionName,
  getJwtSecret,
  getTtlSec,
  getMaxBytes,
  getPlatformCredential,
};
// 文件: cloudfunctions/songjiang-api/lib/http.js
const { getGameKey } = require('./config');
const CORS_HEADERS = {
  'Access-Control-Allow-Origin': '*',
  'Access-Control-Allow-Methods': 'GET,POST,OPTIONS',
  'Access-Control-Allow-Headers': 'Content-Type,Authorization',
  'Access-Control-Max-Age': '86400',
};
function respond(statusCode, body, extraHeaders = {}) {
  return {
    statusCode,
    headers: {
      'Content-Type': 'application/json; charset=utf-8',
      ...CORS_HEADERS,
      ...extraHeaders,
    },
    body: typeof body === 'string' ? body : JSON.stringify(body),
    isBase64Encoded: false,
  };
}
function preflight() {
  return {
    statusCode: 204,
    headers: { ...CORS_HEADERS },
    body: '',
    isBase64Encoded: false,
  };
```

```
}
function parseEvent(event) {
  event = event || {};
  if (event.httpMethod) {
    const method = String(event.httpMethod).toUpperCase();
    let rawBody = event.body || '';
    if (event.isBase64Encoded && rawBody) {
      try {
        rawBody = Buffer.from(rawBody, 'base64').toString('utf8');
      } catch (_) {}
    }
    let body = {};
    if (rawBody) {
      try {
        body = JSON.parse(rawBody);
      } catch (_) {
        body = {};
      }
    }
    return {
      method,
      path: normalizePath(event.path || '/'),
      body,
      headers: lowercaseHeaders(event.headers || {}),
      query: event.queryStringParameters || {},
      raw: event,
    };
  }
  const action = String(event.action || '').replace(/^\/+/, '');
  return {
    method: 'POST',
    path: action ? `/${action}` : '/',
    body: event.body || {},
    headers: lowercaseHeaders(event.headers || {}),
    query: {},
    raw: event,
  };
}
function normalizePath(path) {
  if (!path) return '/';
  let p = String(path);
  if (!p.startsWith('/')) p = `/${p}`;
  const apiPrefix = `${getGameKey()}-api`;
  p = p.replace(new RegExp(`^/(?:${escapeRegExp(apiPrefix)})(?=/|$)`), '');
  if (p === '') p = '/';
  if (p.length > 1 && p.endsWith('/')) p = p.slice(0, -1);
  return p;
}
function lowercaseHeaders(headers) {
  const out = {};
  for (const key of Object.keys(headers || {})) {
    out[key.toLowerCase()] = headers[key];
```

```
  }
  return out;
}
function httpError(status, code, message) {
  const err = new Error(message || code);
  err.status = status;
  err.code = code;
  return err;
}
function escapeRegExp(text) {
  return String(text).replace(/[.*+?^${}()|[\]\\]/g, '\\$&');
}
module.exports = {
  respond,
  preflight,
  parseEvent,
  httpError,
};
// 文件: cloudfunctions/songjiang-api/lib/db.js
const tcb = require('@cloudbase/node-sdk');
const { getCollectionName } = require('./config');
let app = null;
function getApp() {
  if (app) return app;
  app = tcb.init({
    env: process.env.TCB_ENV || tcb.SYMBOL_CURRENT_ENV,
  });
  return app;
}
function getCollection() {
  return getApp().database().collection(getCollectionName('playerData'));
}
module.exports = {
  getApp,
  getCollection,
};
// 文件: cloudfunctions/songjiang-api/lib/geo.js
const IP2Region = require('ip2region').default;
/**
 * 按请求 IP 定省份。ip2region 离线库，IPv4 / IPv6 都能查，不出网。
 * 查不到（内网、海外、头里没 IP）返回空串，由调用方决定退回什么。
 */
const REGIONS = new Set([
  '北京', '天津', '河北', '山西', '内蒙古', '辽宁', '吉林', '黑龙江', '上海', '江苏', '浙江', '安徽',
  '福建', '江西', '山东', '河南', '湖北', '湖南', '广东', '广西', '海南', '重庆', '四川', '贵州',
  '云南', '西藏', '陕西', '甘肃', '青海', '宁夏', '新疆', '台湾', '香港', '澳门', '江湖',
]);
let query = null;
function db() {
  if (!query) query = new IP2Region();
  return query;
}
```

```
function clientIp(req) {
  const h = (req && req.headers) || {};
  const raw = h['x-forwarded-for'] || h['x-real-ip'] || h['x-client-ip'] || '';
  let ip = typeof raw === 'string' ? raw.split(',')[0].trim() : '';
  if (!ip) {
    const ctx = req && req.raw && req.raw.requestContext;
    ip = (ctx && ctx.sourceIp) || '';
  }
  return ip.replace(/^::ffff:/, '');
}
function province(name) {
  return String(name || '')
    .replace(/(维吾尔|壮族|回族)?自治区$/, '')
    .replace(/特别行政区$/, '')
    .replace(/[省市]$/, '')
    .trim();
}
function regionOfIp(ip) {
  if (!ip) return '';
  let hit;
  try {
    hit = db().search(ip);
  } catch (_) {
    return '';
  }
  if (!hit) return '';
  const country = province(hit.country);
  if (REGIONS.has(country)) return country;
  if (country !== '中国') return '';
  const p = province(hit.province);
  return REGIONS.has(p) ? p : '';
}
function regionOf(req) {
  return regionOfIp(clientIp(req));
}
module.exports = {
  REGIONS,
  clientIp,
  regionOf,
  regionOfIp,
};
// 文件: cloudfunctions/songjiang-api/lib/auth.js
const jwt = require('jsonwebtoken');
const { httpError } = require('./http');
const {
  getGameKey,
  gameKeyUpper,
  getJwtSecret: readJwtSecret,
  getTtlSec,
  getPlatformCredential,
} = require('./config');
const SUPPORTED_PLATFORMS = new Set(['wx', 'dy', 'anon']);
```

```
function getJwtSecret() {
  const secret = readJwtSecret();
  if (!secret) {
    throw httpError(500, 'NO_JWT_SECRET', `${gameKeyUpper()}_JWT_SECRET 未配置`);
  }
  return secret;
}
async function handleLogin(req) {
  const body = req.body || {};
  const platform = String(body.platform || '').toLowerCase();
  if (!SUPPORTED_PLATFORMS.has(platform)) {
    throw httpError(400, 'BAD_PLATFORM', `unsupported platform: ${platform}`);
  }
  let platformUid = '';
  if (platform === 'wx') {
    platformUid = (await wxCode2Openid(body.code)).openid;
  } else if (platform === 'dy') {
    platformUid = await ttCode2Openid(body.code);
  } else if (platform === 'anon') {
    const id = String(body.anonId || '').trim();
    if (!id) throw httpError(400, 'NO_ANON_ID', 'anonId 缺失');
    if (!/^[A-Za-z0-9_\-:.]{8,128}$/.test(id)) {
      throw httpError(400, 'BAD_ANON_ID', 'anonId 非法');
    }
    platformUid = id;
  }
  const userId = `${platform}:${platformUid}`;
  const ttlSec = getTtlSec();
  const gameKey = getGameKey();
  const now = Math.floor(Date.now() / 1000);
  const token = jwt.sign(
    { sub: userId, plt: platform, gk: gameKey, iat: now },
    getJwtSecret(),
    { expiresIn: ttlSec },
  );
  return {
    token,
    userId,
    platform,
    gameKey,
    expiresAt: (now + ttlSec) * 1000,
    ttlSec,
  };
}
function requireUser(req) {
  const authHeader = (req.headers && req.headers.authorization) || '';
  const match = /^Bearer\s+(.+)$/i.exec(authHeader);
  if (!match) throw httpError(401, 'NO_TOKEN', '缺少 Authorization: Bearer <token>');
  let payload;
  try {
    payload = jwt.verify(match[1].trim(), getJwtSecret());
  } catch (error) {
```

```
    throw httpError(401, 'BAD_TOKEN', error && error.message ? error.message : 'token 无效');
  }
  const userId = payload && payload.sub;
  if (!userId || typeof userId !== 'string' || !userId.includes(':')) {
    throw httpError(401, 'BAD_TOKEN', 'token sub 非法');
  }
  const currentGameKey = getGameKey();
  if (payload.gk && payload.gk !== currentGameKey) {
    throw httpError(401, 'BAD_TOKEN', `token gameKey=${payload.gk} 与当前 GAME_KEY=${currentGameKey} 不匹配`);
  }
  return { userId, platform: payload.plt || userId.split(':')[0] };
}
async function wxCode2Openid(code) {
  const appid = getPlatformCredential('wx', 'APPID');
  const secret = getPlatformCredential('wx', 'SECRET');
  if (!appid || !secret) throw httpError(500, 'NO_WX_CFG', `${gameKeyUpper()}_WX_APPID/${gameKeyUpper()}_WX_SECRET
    未配置`);
  if (!code) throw httpError(400, 'NO_CODE', 'wx code 缺失');
  const url = `https://api.weixin.qq.com/sns/jscode2session?appid=${encodeURIComponent(appid)}&secret=${encodeURICompon
    ent(secret)}&js_code=${encodeURIComponent(code)}&grant_type=authorization_code`;
  const data = await httpGetJson(url);
  if (!data || !data.openid) {
    throw httpError(401, 'WX_LOGIN_FAIL', `wx code2session 失败: ${JSON.stringify(data || {})}`);
  }
  return { openid: data.openid, sessionKey: data.session_key || '' };
}
async function ttCode2Openid(code) {
  const appid = getPlatformCredential('tt', 'APPID');
  const secret = getPlatformCredential('tt', 'SECRET');
  if (!appid || !secret) throw httpError(500, 'NO_TT_CFG', `${gameKeyUpper()}_TT_APPID/${gameKeyUpper()}_TT_SECRET 未
    配置`);
  if (!code) throw httpError(400, 'NO_CODE', 'dy code 缺失');
  const data = await httpPostJson('https://developer.toutiao.com/api/apps/v2/jscode2session', { appid, secret, code });
  if (!data || data.err_no !== 0 || !data.data || !data.data.openid) {
    throw httpError(401, 'TT_LOGIN_FAIL', `dy code2session 失败: ${JSON.stringify(data || {})}`);
  }
  return data.data.openid;
}
function httpGetJson(url) {
  return httpRequestJson(url, 'GET');
}
function httpPostJson(url, body) {
  return httpRequestJson(url, 'POST', body);
}
function httpRequestJson(url, method, body) {
  if (typeof fetch !== 'function') {
    return Promise.reject(httpError(500, 'NO_FETCH', '当前 Node 运行时不支持 fetch，请使用 Node 18+'));
  }
  const opts = { method, headers: { 'Content-Type': 'application/json' } };
  if (body !== undefined) opts.body = JSON.stringify(body);
  return fetch(url, opts).then(async (res) => {
    const text = await res.text();
```

```
    try {
      return JSON.parse(text);
    } catch (_) {
      return { _raw: text };
    }
  });
}
module.exports = {
  handleLogin,
  requireUser,
};
// 文件: cloudfunctions/songjiang-api/lib/live.js
const { httpError } = require('./http');
const { requireUser } = require('./auth');
const { getApp } = require('./db');
const { getCollectionName } = require('./config');
const { REGIONS, regionOf } = require('./geo');
/**
 * 运营配置：集合 songjiang_config 里 _id = "live" 的一条文档，没有就用默认值。
 *   { noticeTitle, notice, noticeVer, adUnitId, adCaps: [{ scene, cap }] }
 * 排行榜：集合 songjiang_rank，一人一条，按 userId 唯一。
 * 地区不让玩家选，按请求 IP 定省份；查不到时沿用上次的，再没有就是「江湖」。
 */
const DEFAULT_CAPS = [
  { scene: 'stamina', cap: 8 },
  { scene: 'double_coin', cap: 10 },
  { scene: 'keep_star', cap: 5 },
  { scene: 'free_item', cap: 6 },
];
const MAX_SCORE = 100000;
const LIST_LIMIT = 50;
function db() {
  return getApp().database();
}
function shanghaiDay(ms) {
  const d = new Date(ms + 8 * 3600 * 1000);
  const y = d.getUTCFullYear();
  const m = String(d.getUTCMonth() + 1).padStart(2, '0');
  const day = String(d.getUTCDate()).padStart(2, '0');
  return `${y}${m}${day}`;
}
async function readLive() {
  try {
    const res = await db().collection(getCollectionName('config')).doc('live').get();
    const doc = res && Array.isArray(res.data) ? res.data[0] : res && res.data;
    return doc || {};
  } catch (_) {
    return {};
  }
}
async function handleConfig(req) {
  const now = Date.now();
```

```
  const live = await readLive();
  const caps = Array.isArray(live.adCaps) && live.adCaps.length > 0 ? live.adCaps : DEFAULT_CAPS;
  return {
    serverNow: now,
    day: shanghaiDay(now),
    noticeTitle: String(live.noticeTitle || ''),
    notice: String(live.notice || ''),
    noticeVer: Number(live.noticeVer) || 0,
    adUnitId: String(live.adUnitId || ''),
    adCaps: caps
      .filter((c) => c && typeof c.scene === 'string')
      .map((c) => ({ scene: c.scene, cap: Math.max(0, Math.floor(Number(c.cap) || 0)) })),
    region: regionOf(req),
  };
}
function cleanName(n) {
  const v = Array.from(String(n || '').replace(/[\u0000-\u001f<>]/g, '').trim()).slice(0, 12).join('');
  return v || '好汉';
}
function cleanAvatar(u) {
  const v = String(u || '').trim();
  if (!/^https:\/\/\S{8,2000}$/.test(v)) return '';
  return v;
}
async function findMine(col, userId) {
  const res = await col.where({ userId }).limit(1).get();
  return res && Array.isArray(res.data) ? res.data[0] : null;
}
function regionFor(req, row) {
  const seen = regionOf(req);
  if (seen) return seen;
  if (row && REGIONS.has(row.region)) return row.region;
  return '江湖';
}
async function handleRankSubmit(req) {
  const { userId } = requireUser(req);
  const body = req.body || {};
  const score = Math.floor(Number(body.score));
  if (!Number.isFinite(score) || score < 0 || score > MAX_SCORE) {
    throw httpError(400, 'BAD_SCORE', 'score 非法');
  }
  const col = db().collection(getCollectionName('rank'));
  const row = await findMine(col, userId);
  const region = regionFor(req, row);
  const doc = {
    userId,
    name: cleanName(body.name),
    avatar: cleanAvatar(body.avatar) || (row && cleanAvatar(row.avatar)) || '',
    region,
    score,
    tier: Math.max(0, Math.floor(Number(body.tier) || 0)),
    stars: Math.max(0, Math.floor(Number(body.stars) || 0)),
```

```
    wins: Math.max(0, Math.floor(Number(body.wins) || 0)),
    updatedAt: Date.now(),
  };
  if (row && row._id) await col.doc(row._id).update(doc);
  else await col.add(doc);
  return { ok: true, region };
}
/** body.region 传「全国」拿总榜，其它都按自己所在的省拿省榜。 */
async function handleRankList(req) {
  const { userId } = requireUser(req);
  const body = req.body || {};
  const col = db().collection(getCollectionName('rank'));
  const _ = db().command;
  let me = await findMine(col, userId);
  const myRegion = regionFor(req, me);
  if (me && me._id && me.region !== myRegion) {
    await col.doc(me._id).update({ region: myRegion });
    me = { ...me, region: myRegion };
  }
  const all = String(body.region || '') === '全国';
  const filter = all ? {} : { region: myRegion };
  const topRes = await col.where(filter).orderBy('score', 'desc').orderBy('updatedAt', 'asc').limit(LIST_LIMIT).get();
  const top = (topRes && topRes.data ? topRes.data : []).map((r) => ({
    name: r.name,
    avatar: cleanAvatar(r.avatar),
    region: r.region,
    score: r.score,
    tier: r.tier,
    stars: r.stars,
    me: r.userId === userId,
  }));
  let rank = 0;
  if (me) {
    const at = top.findIndex((r) => r.me);
    if (at >= 0) {
      // 人就在这页里时，名次就是列表里的位置，底下「我」和上面对得上。
      rank = at + 1;
    } else {
      // 落到 50 名开外：分高的在前，同分按先上报的在前，和 orderBy 一致。
      const higher = await col.where({ ...filter, score: _.gt(me.score) }).count();
      const earlier = await col.where({ ...filter, score: me.score, updatedAt: _.lt(Number(me.updatedAt) || 0) }).count();
      rank = ((higher && higher.total) || 0) + ((earlier && earlier.total) || 0) + 1;
    }
  }
  const count = await col.where(filter).count();
  return {
    region: all ? '全国' : myRegion,
    myRegion,
    top,
    rank,
    total: (count && count.total) || 0,
  };
```

```
}
const https = require('https');
const { URL } = require('url');
// 真机的 request 域名里没有微信头像域名，客户端改走这里中转。
const AVATAR_HOSTS = new Set(['thirdwx.qlogo.cn', 'wx.qlogo.cn', 'mmbiz.qpic.cn', 'thirdqq.qlogo.cn']);
const AVATAR_MAX = 262144;
function avatarUrlOf(raw) {
  let u;
  try { u = new URL(String(raw || '').trim()); } catch (_) { return null; }
  if (u.protocol !== 'https:' || !AVATAR_HOSTS.has(u.hostname)) return null;
  return u;
}
function isImage(buf) {
  if (!buf || buf.length < 8) return false;
  if (buf[0] === 0xff && buf[1] === 0xd8) return true;
  if (buf[0] === 0x89 && buf[1] === 0x50 && buf[2] === 0x4e && buf[3] === 0x47) return true;
  if (buf[0] === 0x47 && buf[1] === 0x49 && buf[2] === 0x46) return true;
  return false;
}
function readAvatar(url, redirects) {
  return new Promise((resolve, reject) => {
    const u = avatarUrlOf(url);
    if (!u) {
      reject(httpError(400, 'BAD_AVATAR', '头像地址不对'));
      return;
    }
    const req = https.get(u, { timeout: 8000, headers: { 'User-Agent': 'songjiang' } }, (res) => {
      const code = res.statusCode || 0;
      if (code >= 300 && code < 400 && res.headers.location && redirects < 2) {
        res.resume();
        let next = '';
        try { next = new URL(res.headers.location, u).toString(); } catch (_) { next = ''; }
        resolve(readAvatar(next, redirects + 1));
        return;
      }
      if (code !== 200) {
        res.resume();
        reject(httpError(502, 'AVATAR_FETCH', '头像下不来'));
        return;
      }
      const chunks = [];
      let size = 0;
      let failed = false;
      res.on('data', (c) => {
        if (failed) return;
        size += c.length;
        if (size > AVATAR_MAX) {
          failed = true;
          req.destroy();
          reject(httpError(413, 'AVATAR_BIG', '头像太大'));
          return;
        }
```

```
      chunks.push(c);
    });
    res.on('end', () => {
      if (failed) return;
      const buf = Buffer.concat(chunks);
      if (!isImage(buf)) {
        reject(httpError(502, 'AVATAR_FETCH', '头像下不来'));
        return;
      }
      resolve(buf);
    });
  });
  req.on('error', () => reject(httpError(502, 'AVATAR_FETCH', '头像下不来')));
  req.on('timeout', () => {
    req.destroy();
    reject(httpError(502, 'AVATAR_FETCH', '头像下不来'));
  });
 });
}
async function handleAvatar(req) {
 requireUser(req);
 const buf = await readAvatar(req.body && req.body.url, 0);
 return { b64: buf.toString('base64') };
}
module.exports = {
 handleConfig,
 handleRankSubmit,
 handleRankList,
 handleAvatar,
 shanghaiDay,
};
// 文件: cloudfunctions/songjiang-api/lib/save.js
const { httpError } = require('./http');
const { requireUser } = require('./auth');
const { getCollection } = require('./db');
const { getMaxBytes } = require('./config');
async function handlePull(req) {
 const { userId, platform } = requireUser(req);
 const col = getCollection();
 const res = await col.where({ userId }).limit(1).get();
 const doc = (res && Array.isArray(res.data) && res.data[0]) || null;
 if (!doc) {
  return {
    userId,
    platform,
    exists: false,
    schemaVersion: 0,
    updatedAt: 0,
    payload: {},
    payloadKeys: [],
  };
 }
```

```
  return {
    userId,
    platform,
    exists: true,
    schemaVersion: doc.schemaVersion || 0,
    updatedAt: doc.updatedAt || 0,
    payload: doc.payload || {},
    payloadKeys: Array.isArray(doc.payloadKeys) ? doc.payloadKeys : Object.keys(doc.payload || {}),
    clientFingerprint: doc.clientFingerprint || '',
  };
}
async function handlePush(req) {
  const { userId, platform } = requireUser(req);
  const body = req.body || {};
  const schemaVersion = Number(body.schemaVersion);
  const updatedAt = Number(body.updatedAt);
  const baseRemoteUpdatedAt = Number(body.baseRemoteUpdatedAt || 0);
  const payload = body.payload;
  const clientFingerprint = String(body.clientFingerprint || '').slice(0, 200);
  if (!Number.isFinite(schemaVersion) || schemaVersion <= 0) {
    throw httpError(400, 'BAD_SCHEMA', 'schemaVersion 非法');
  }
  if (!Number.isFinite(updatedAt) || updatedAt <= 0) {
    throw httpError(400, 'BAD_UPDATED_AT', 'updatedAt 非法');
  }
  if (!Number.isFinite(baseRemoteUpdatedAt) || baseRemoteUpdatedAt < 0) {
    throw httpError(400, 'BAD_BASE_REMOTE_UPDATED_AT', 'baseRemoteUpdatedAt 非法');
  }
  if (!payload || typeof payload !== 'object' || Array.isArray(payload)) {
    throw httpError(400, 'BAD_PAYLOAD', 'payload 必须是 object');
  }
  const size = Buffer.byteLength(JSON.stringify(payload), 'utf8');
  const maxBytes = getMaxBytes();
  if (size > maxBytes) {
    throw httpError(413, 'PAYLOAD_TOO_LARGE', `payload 超限: ${size}B > ${maxBytes}B`);
  }
  const payloadKeys = [];
  for (const k of Object.keys(payload)) {
    if (typeof payload[k] !== 'string') {
      throw httpError(400, 'BAD_PAYLOAD_VALUE', `payload[${k}] 必须是字符串`);
    }
    payloadKeys.push(k);
  }
  const col = getCollection();
  const existingRes = await col.where({ userId }).limit(1).get();
  const existing = (existingRes && Array.isArray(existingRes.data) && existingRes.data[0]) || null;
  if (existing) {
    const prevUpdatedAt = Number(existing.updatedAt) || 0;
    if (updatedAt < prevUpdatedAt || baseRemoteUpdatedAt < prevUpdatedAt) {
      throw Object.assign(
        httpError(409, 'STALE_UPDATE', `服务端已有更新版本 remote=${prevUpdatedAt} > local=${updatedAt}, base=${baseRe
          moteUpdatedAt}`),
```

```
      {
        data: {
          remote: {
            schemaVersion: existing.schemaVersion || 0,
            updatedAt: prevUpdatedAt,
            payload: existing.payload || {},
            payloadKeys: Array.isArray(existing.payloadKeys)
              ? existing.payloadKeys
              : Object.keys(existing.payload || {}),
          },
        },
      },
    );
  }
}
  const mergedPayload = { ...(existing && existing.payload ? existing.payload : {}) };
  for (const k of Object.keys(payload)) mergedPayload[k] = payload[k];
  const mergedPayloadKeys = Object.keys(mergedPayload);
  const mergedSize = Buffer.byteLength(JSON.stringify(mergedPayload), 'utf8');
  if (mergedSize > maxBytes) {
    throw httpError(413, 'PAYLOAD_TOO_LARGE', `payload 超限: ${mergedSize}B > ${maxBytes}B`);
  }
  const now = Date.now();
  const docData = {
    userId,
    platform,
    schemaVersion,
    updatedAt,
    baseRemoteUpdatedAt,
    clientFingerprint,
    payload: mergedPayload,
    payloadKeys: mergedPayloadKeys,
    lastWriteAt: now,
  };
  if (existing && existing._id) {
    await col.doc(existing._id).update(docData);
    return { userId, updatedAt, savedAt: now, mode: 'update', sizeBytes: mergedSize };
  }
  const addRes = await col.add(docData);
  return {
    userId,
    updatedAt,
    savedAt: now,
    mode: 'insert',
    sizeBytes: size,
    _id: addRes && (addRes.id || addRes._id),
  };
}
module.exports = {
  handlePull,
  handlePush,
};
```